

Dynamics of Expenditures on Durable Goods: the Role of New-Product Quality*

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Abstract

We study the role of new-product quality for the dynamics of durable-goods expenditures around the Great Recession. We assemble a rich dataset on US new-car markets during 2004-2012, combining data on transaction prices with detailed information about vehicles' technical characteristics. During the recession, a reallocation of expenditures away from high-quality new models accounts for a significant decline in the dispersion of expenditures. In turn, car manufacturers introduced new models of lower quality. The drop in new-model quality persistently depressed the technology embodied in vehicles, and likely contributed to the slow recovery of expenditures.

JEL Codes: E21, E23, E32.

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1 Introduction

Households adopt new technologies by purchasing new durable goods, such as vehicles. During the Great Recession of 2008-2009, consumer expenditures on durable goods dropped by approximately 17%. Expenditures on motor vehicles—which constitute approximately 35% of durable-goods expenditures—accounted for more than half of this decrease and remained low during the recovery.

The goal of this paper is to empirically investigate the role of new-product quality for these dynamics. Our descriptive analysis suggests that complementary demand and supply factors contributed to a downward quality adjustment in durable-goods purchases during the recession. Specifically, households reallocated their purchases of new cars toward cheaper models—which tend to be continuing models, of lower quality than new models—or delayed their purchases. Amid this decline in demand, manufacturers introduced new models of low quality, persistently depressing the path of technology.

Cars represent an ideal object for our analysis for two reasons. First, they are a large and procyclical component of durable-goods expenditures. Second, detailed information on car markets allows us to measure quality dynamics, providing evidence on the importance of new products. To this end, we assemble a rich dataset on US new-car markets, combining two data sources. The first dataset contains the universe of new-car transactions in several US states between 2004 and 2012 and reports transaction prices as well as car features, such as make and model. The second dataset contains detailed information on the technical characteristics of each vehicle model sold in the US during the same period.

We exploit these data to provide new evidence about the distribution of vehicle expenditures and quality around the Great Recession. Our analysis proceeds in four steps, each yielding a main finding.

First, we document a drop in the dispersion of new-car expenditures during the Great Recession—and a smaller decline in the average price—due to a decline in the volume of high-price transactions.

Second, we show that the drop in dispersion is due to expenditure reallocation across models—specifically, a decline in expenditures on expensive newly introduced models. Furthermore, exploiting geographical variation, we relate this drop in demand for high-quality models with the severity of the recession. Because the supply of vehicles does not vary across locations, this finding shows that shocks to household demand play a primary role in the downward quality adjustment.

Third, we connect car prices and characteristics. We use hedonic regressions to construct

a measure of vertical quality that summarizes vehicle technical characteristics ([Griliches, 1961](#)). We show that compositional changes in the characteristics of cars sold account for the drop in expenditures. Furthermore, vehicle quality, based on pre-recession hedonic prices, displays no growth during the recovery.

Fourth, we estimate the level of technology embodied in vehicles using only data on car characteristics. We document that new models introduced during the Great Recession featured a significantly worse trade-off between their main attributes than models introduced in other years. This finding is consistent with an endogenous response of manufacturers that contributed to the drop in durable-goods quality. Moreover, this technological slowdown had persistent effects throughout the recovery, reducing the quality of the stock of registered vehicles.

Overall, our analysis highlights the complementary role of demand and supply forces for quality dynamics. The narrative that emerges from our findings is that a drop in household demand for quality led to an endogenous response on the supply side, with a decrease in both volume and quality of new products, which further reduced technology adoption.

Our findings have several implications. Most directly, the motor vehicle industry experienced a deep crisis in 2008-2009, which led to a drop in employment and government bailouts. Moreover, because of the centrality of this industry in the US production structure, the effects of this crisis spread across different sectors.¹ Thus, understanding the micro dynamics of expenditures on vehicles is an important step toward understanding the Great Recession and the subsequent slow recovery.

Furthermore, our findings have broader implications, contributing to several strands of the literature. First, we provide evidence for the complementary roles of demand and supply factors for innovation and technology adoption. Several papers show that downward adjustment in consumer demand for quality is an important margin in the Great Recession ([Jaimovich, Rebelo, and Wong, 2019](#); [Argente and Lee, 2021](#)).² A related literature emphasizes the entry and exit of retail products as an important margin for the evolution of technology around the same period ([Argente, Lee, and Moreira, 2018](#); [Jaravel, 2019](#); [Granja and Moreira, 2020](#)).³ Whereas these studies mainly focus on services and nondurable goods, we analyze one of the most important household durable goods, building on the insights

¹[Atalay \(2017\)](#) and [vom Lehn and Winberry \(2022\)](#) document that the auto industry plays a central role in the US production network.

²[Fisher, Johnson, and Smeeding \(2013\)](#) and [Meyer and Sullivan \(2013\)](#) find that consumption inequality declined during the Great Recession.

³[Broda and Weinstein \(2010\)](#) documents that product creation is procyclical during the period 1999-2003.

of [Bils and Klenow \(2001\)](#) and [Bils \(2009\)](#). The evidence on complementarity between demand and supply is also consistent with the mechanism in [Shleifer \(1986\)](#).⁴

Second, a large literature studies the role of durable goods for business cycles (for seminal contributions, see [Mankiw, 1982](#); [Bernanke, 1985](#); [Caballero, 1993](#)). An important force in models of durables demand ([Barsky, House, and Kimball, 2007](#); [Berger and Vavra, 2015](#); [Dapor, Li, Mehkari, and Tsai, 2018](#); [Attanasio, Larkin, Ravn, and Padula, 2020](#); [Gavazza and Lanteri, 2021](#); [McKay and Wieland, 2021](#); [Beraja and Wolf, 2022](#)) is intertemporal substitution, which implies that pent-up demand may induce strong recoveries after drops in expenditures on durables.⁵ However, expenditures on durables recovered sluggishly after the Great Recession. Our finding that new-car quality was persistently depressed during the recovery may partially account for the slow recovery in expenditures.

Relatedly, our findings on the persistent implications of the downward quality adjustment during the Great Recession are consistent with the literature on medium-run business cycles (e.g., [Comin and Gertler, 2006](#); [Benigno and Fornaro, 2018](#); [Anzoategui, Comin, Gertler, and Martinez, 2019](#); [Bianchi, Kung, and Morales, 2019](#); [Vinci and Licandro, 2020](#)). Our contribution is to measure the medium-run effects of new-product introduction around the Great Recession.

2 Data

Our empirical analysis exploits two datasets on new-car transactions and model characteristics, respectively. We introduce them in this section.

New-car Prices, [Dominion Dealer Solutions \(2019\)](#). This dataset (henceforth Dominion dataset) reports the universe of new-car sales in five states—Colorado, Idaho, North Dakota, Ohio, and Texas—for the period 2004-2012. For each sale, the dataset reports the transaction price, the month of the transaction, and the make, model, body type, and trim of the vehicle. The dataset contains more than 16.5 million vehicle transactions.⁶

⁴[Acemoglu and Linn \(2004\)](#) provides related evidence from the pharmaceutical industry and [Einav \(2007\)](#) from the motion picture industry.

⁵Several papers build on [Eberly \(1994\)](#) and [Attanasio \(2000\)](#), which abstract from business cycles. [Adda and Cooper \(2000, 2006\)](#) and [Gavazza, Lizzeri, and Roketskiy \(2014\)](#) develop quantitative models of car replacement.

⁶For North Dakota, prices are reported for 2008-2012 only. Transaction prices in Colorado exhibit some unusually low values than those in the other states; all our empirical findings are robust to excluding transactions in Colorado.

New-car Model Characteristics, IHS Markit (2020). This dataset (henceforth IHS dataset) reports detailed characteristics of all new passenger-car models sold during 2003–2012, including make, model, trim, body type, generation year, dimensions, as well as engine attributes, such as size and horsepower, fuel type, fuel consumption, transmission, and turbo injection.⁷

The dataset also reports the aggregate number of US sales for each model at annual frequency during 2003–2012. We exclude pick-up trucks from our analysis because the dataset does not have comprehensive information about them.

The product life cycle of cars typically features the replacement of a “generation” of a car model with a new generation on average every 5.8 years. For example, all 2007–2011 Toyota Camry models belong to the 2007 generation. Whereas small changes in characteristics happen at annual frequency within a generation, a new generation features a larger redesign. Hence, we define a vehicle model in the IHS data as a triplet of make, model, and generation. We further define a new model in year t as a model for which we observe the first transaction in year t or $t - 1$, to account for the fact that the first transaction on a new model tends to appear in the second half of the year. This definition of a new model encompasses entirely new model names.

Based on this definition, we merge the Dominion and IHS datasets by matching vehicle models across the two datasets and allocating each transaction in the Dominion dataset to a model generation in the IHS dataset. Appendix A provides more details on our model definition and procedure to merge the datasets.

We thus obtain a rich dataset on car sales that combines information on prices and technical characteristics. Throughout the paper, we refer to a car model as a make-model-generation triplet. According to this industry-wide definition, our dataset contains over 500 models.

3 Empirical Patterns

In this section, we describe several empirical patterns: (i) we document the dynamics of the distribution of expenditures on new cars around the Great Recession; (ii) we decompose the dispersion in expenditures highlighting the role of new models; (iii) we relate expendi-

⁷Information about weight is missing in approximately 40% of models. Thus, we use all models for which we observe their weight to estimate a log-linear relationship between weight and other physical dimensions: wheelbase, width, height, and number of seats. This regression has an R^2 of 0.93. We use its predicted values to impute the weight whenever we do not observe it.

tures to car characteristics; and (iv) we analyze the level of technology embodied in cars. Appendix B reports additional details and robustness checks.

3.1 Dynamics of the Distribution of Expenditures on New Vehicles

We begin by describing the distribution of expenditures on new cars in the Dominion dataset. Figure 1 displays the main features of this distribution during 2004-2012. The transactions in this dataset provide a representative account of the dramatic effects of the Great Recession on US car markets: The top-left panel shows that the total number of new-car sales drops by approximately 30% during the recession and only returns to pre-recession levels in 2012, similar to the US aggregate dynamics.

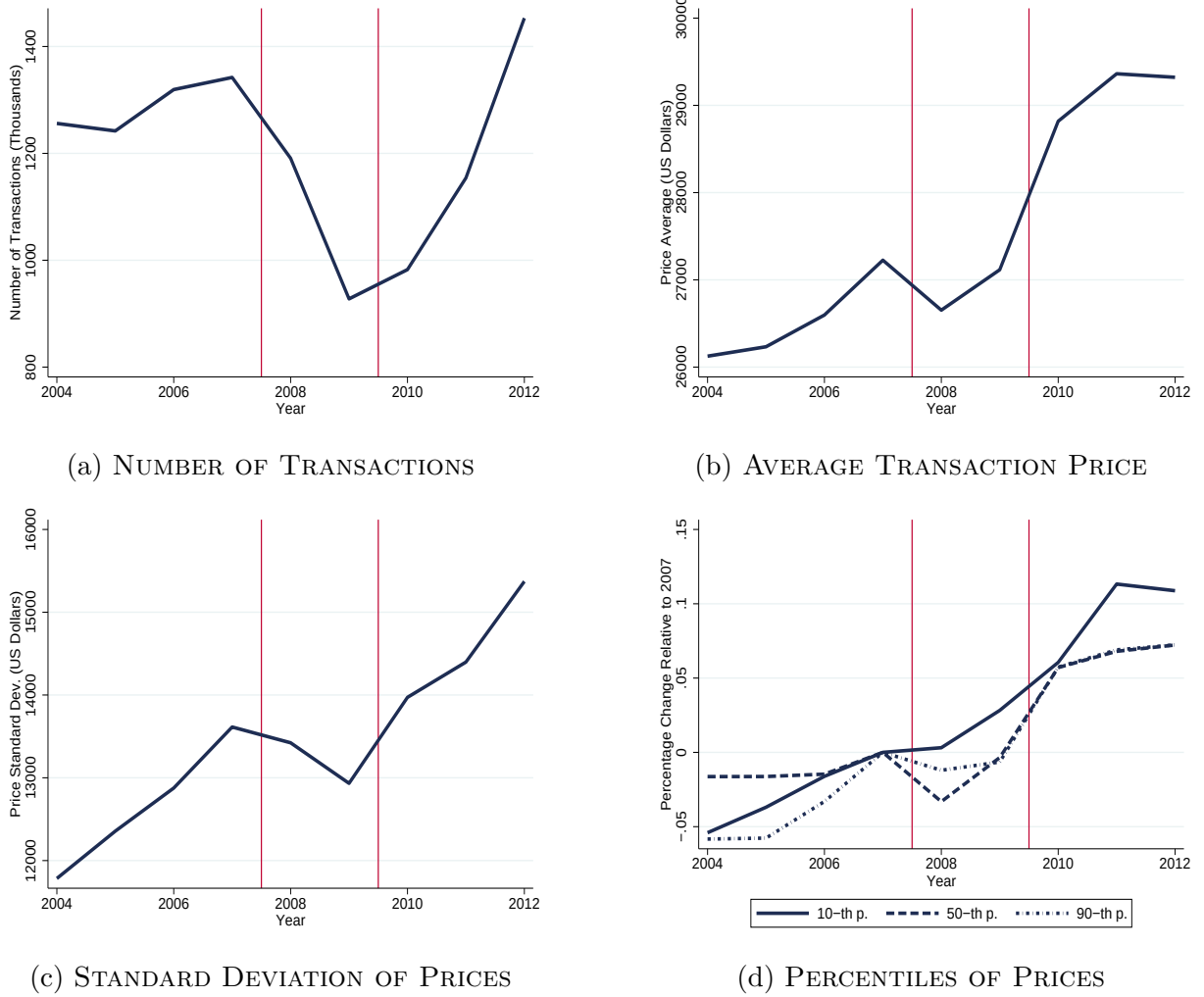
We thus exploit the dataset to analyze the micro dynamics of the expenditure distribution. The top-right panel plots the average transaction price; the bottom-left panel the standard deviation; and the bottom-right panel the 10th, 50th, and 90th percentiles of the distribution, all normalized to zero in 2007.

Both first and second moments of the expenditure distribution display an increasing trend. On average, transaction prices increase by 1.6% annually between 2004 and 2012. However, during the Great Recession, we observe a decline in the average price and a larger decline in the dispersion of prices. Notably, the average price, which equals \$27,226 in 2007, displays a peak-to-trough decline of approximately 2%. The standard deviation, which equals \$13,614 in 2007, declines by approximately 5%. Relative to their respective trends, the average price drops by approximately 3% and the standard deviation by approximately 6% during the recession. In summary, our first main finding is that the decrease in dispersion during the recession is about twice as large as the decrease in average expenditures.

The evidence on the first two moments of the distribution suggests that households reallocated their expenditures away from expensive vehicles during the recession. Different percentiles of the distribution confirm this pattern. Consistent with the low-frequency dynamics of average prices, all percentiles increase over time between 2004 and 2012. However, the median and the 90th percentile decline significantly during the recession, both in absolute terms and relative to their trends. In contrast, the 10th percentile remains on its trend. This analysis suggests that a drop in expenditures on intermediate- and high-quality cars accounts for the declines in the average and the dispersion of expenditures.

These findings are consistent with the evidence on household expenditures based on

Figure 1: Dynamics of New-Vehicle Expenditures



Notes: The figure displays the number of new-car sales (top-left panel), the average (top-right panel), the standard deviation (bottom-left panel), and three percentiles—10th, 50th, and 90th—(bottom-right panel) of the distribution of transaction prices from the Dominion dataset. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

the Consumer Expenditure Survey. [Meyer and Sullivan \(2013\)](#) documents a low-frequency increase in consumption inequality and a decrease in dispersion during the Great Recession, with lower percentiles of expenditures displaying smaller declines than higher percentiles. However, our dataset allows us to take further steps to connect the distribution of expenditures with features of the goods purchased.

3.2 Decomposing the Dispersion of Expenditures

We perform several decompositions of the variance of prices to investigate the drivers of the cyclical dynamics of the distribution of expenditures. Our second main finding is that reallocation of expenditures *between* car models—specifically a drop in expenditures on newly introduced models with high price—accounts for the compression in the distribution in the recession. In contrast, average prices conditional on vehicle model do not display significant changes relative to their trend.

3.2.1 Between versus Within Models

We decompose the total variance of expenditures on new vehicles in year t , V_t , as follows:

$$V_t = V_t^B + V_t^W,$$

where V_t^B denotes the between-models component of the total variance and V_t^W denotes the within-model component.⁸ Formally, we have

$$\begin{aligned} V_t &\equiv \frac{1}{N_t} \sum_{i \in M_t} \sum_{j \in X_{it}} (p_{ijt} - \bar{p}_t)^2, \\ V_t^B &\equiv \sum_{i \in M_t} s_{it} (\bar{p}_{it} - \bar{p}_t)^2, \\ V_t^W &\equiv \frac{1}{N_t} \sum_{i \in M_t} \sum_{j \in X_{it}} (p_{ijt} - \bar{p}_{it})^2, \end{aligned}$$

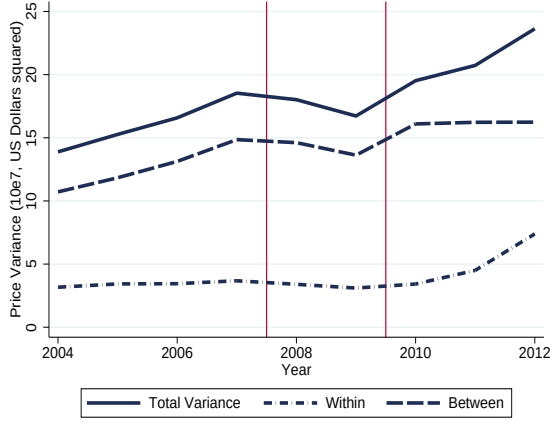
where $i \in M_t$ denotes a model sold in year t , $j \in X_{it}$ denotes a transaction on model i in year t , with market share s_{it} ; N_t is the total number of transactions in year t ; p_{ijt} are individual prices; $\bar{p}_{it}$ is the average price of model i in year t ; and $\bar{p}_t$ is the average price in year t .

The top-left panel of Figure 2 displays the total variance V_t (solid line) and its components: between models V_t^B (dashed line) and within models V_t^W (dashed-dotted line). The between-models component accounts for almost 80% of total variation in prices before the recession, whereas within-model dispersion in transaction prices accounts for approximately 20% of total variation.⁹ Notably, the between-models component accounts for the entire reduction in total dispersion during the recession. In contrast, during the same period

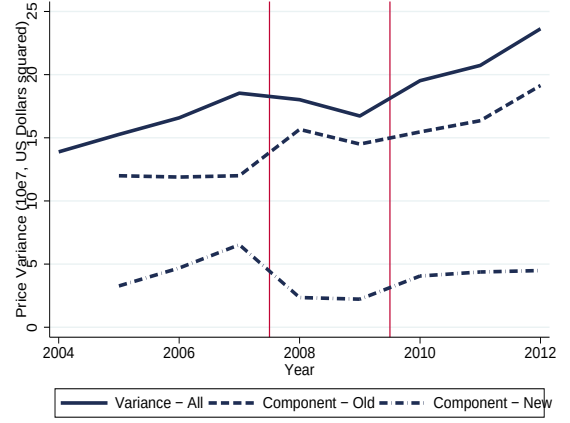
⁸The covariance term equals zero.

⁹Variation in prices within models is mostly due to different trims within each model. This variation does not appear to be relevant for the cyclical dynamics, which confirms that our approach of merging the Dominion and IHS datasets at the model level is sound.

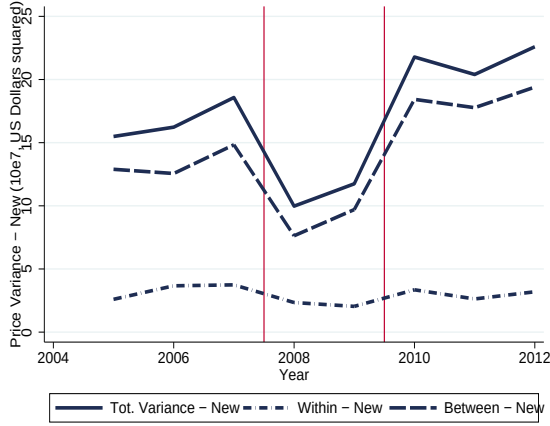
Figure 2: Variance Decomposition



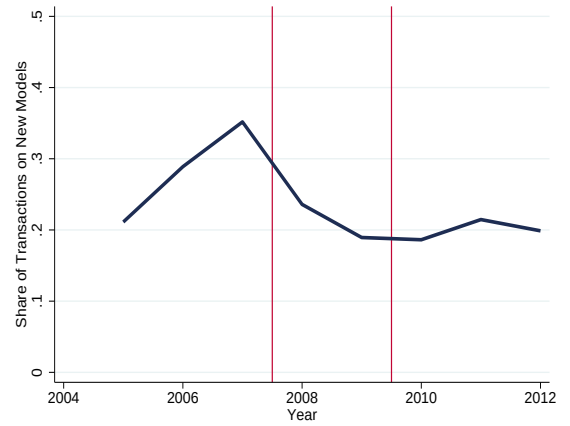
(a) VARIANCE OF NEW-CAR PRICES, BETWEEN AND WITHIN



(b) VARIANCE OF NEW-CAR PRICES, NEW AND CONTINUING



(c) VARIANCE OF NEWLY INTRODUCED MODELS



(d) SHARE OF NEWLY INTRODUCED MODELS

Notes: The figure displays several decompositions of the variance of transaction prices in the Dominion dataset. The top-left panel displays the decomposition of the variance of new-vehicle transaction prices V_t (solid line) into the following components: between models V_t^B (dashed line) and within models V_t^W (dashed-dotted line). The top-right panel displays the decomposition of the variance V_t (solid line) into two components: new models $s_t^N V_t^N$ (dashed-dotted line) and old models $(1 - s_t^N) V_t^O$ (dashed line). The bottom-left panel displays the variance of expenditures on new models V_t^N (solid line) and its decomposition into between-models component $V_t^{N,B}$ (dashed line) and within-models component $V_t^{N,W}$ (dashed-dotted line). The bottom-right panel displays the share of transactions on new models s_t^N . Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

there are no significant changes in the dispersion of prices within models. This evidence establishes that households reallocated their expenditures toward models with a price close to the average.

3.2.2 New versus Old Models

The reallocation of expenditures away from expensive models prompts us to analyze the role of newly introduced models. New models tend to be more expensive than continuing models, fueling the long-run growth in the average price.

Based on our definition of a new model (Section 2), we find that new models play a prominent role in the dynamics of the expenditure distribution. Strikingly, between 2005 and 2007, the average transaction price for new models is \$28,080, which is higher than the average for old models, \$26,144. However, in 2008, the average price of new models drops to \$25,764, which is lower than the average for old models, \$26,927.

We analyze the contribution of new models to the variance of expenditures, using the following variance decomposition:

$$V_t = s_t^N V_t^N + (1 - s_t^N) V_t^O,$$

where s_t^N is the share of transactions on new models in year t and V_t^N (V_t^O) is the variance of expenditures on new (old) models. In turn, these variances equal:

$$\begin{aligned} V_t^N &\equiv \frac{1}{N_t^N} \sum_{i \in M_t^N} \sum_{j \in X_{it}} (p_{ijt} - \bar{p}_t)^2, \\ V_t^O &\equiv \frac{1}{N_t^O} \sum_{i \in M_t^O} \sum_{j \in X_{it}} (p_{ijt} - \bar{p}_t)^2, \end{aligned}$$

where M_t^N and M_t^O are the sets of new and continuing models in year t , and N_t^N and $N_t^O = N_t - N_t^N$ are the respective number of transactions.

The top-right panel of Figure 2 displays the decomposition of the total variance of expenditures V_t into expenditures on new models, $s_t^N V_t^N$, and on old models, $(1 - s_t^N) V_t^O$. The component due to new models displays a sharp drop during the recession, fully accounting for the drop in total variance. This pattern arises for two concurring reasons. First, the dispersion of prices of new models drops by nearly one-half during the recession. The bottom-left panel of Figure 2 portrays the dynamics of the variance of expenditures on new models V_t^N , showing that its between-model component accounts for its drop, consistent with the same decomposition for all models.

Second, the share of transactions on new models s_t^N decreases sharply, from a peak in 2007 of approximately 35% to less than 20% in 2009, as the bottom-right panel of Figure 2 shows, despite the fact that new models were cheaper during the recession.¹⁰ This pattern

¹⁰The 2007 peak in the market share of new models is due to the simultaneous introduction of new generations of three popular models: Toyota Camry, Nissan Altima, and Chevrolet Tahoe. Figure B9 in

suggests a drop in the quality of new models during the recession, which is thus the focus of the following subsections. Nonetheless, we do not observe large changes in the variance of expenditures on old models, V_t^O , relative to its trend, suggesting that households did not substitute the “missing” new models of high quality with old models of high quality—most likely delaying their purchases.

In the aftermath of the recession, the dispersion of expenditures on new models V_t^N returns to its trend. However, Figure 2, as well as Figure B9 in Appendix B, show that neither the share of transactions on new models s_t^N nor the fraction of new models on sale overshoots during the recovery. This evidence suggests that car manufacturers did not simply respond to the recession by delaying the introduction of high-quality new models; rather, there was a missing generation of new products, likely contributing to the slow recovery of expenditures.

In Appendix B, we analyze cross-sectional heterogeneity in new-model introduction across carmakers. We divide carmakers in three groups, depending on their geographical origin (Europe, Asia, and US). This analysis reveals two patterns. First, all groups of carmakers decreased the volume of new-model introduction during the recession. Second, European carmakers specialize in the introduction of high-quality models. As a result, they largely account for the drop in high-quality new models during the recession.

We also decompose the margin of new-model introduction between new model names, which may expand the set of models available to consumers (*horizontal* innovation), and new generations of existing model names, which improve on past generations of existing products (*vertical* innovation). Before the recession, both margins account for approximately half of new-model introduction. During the recession, both margins decline, but the bulk of the overall drop in new models is due to missing new generations of existing models. This evidence (tentatively) suggests that the vertical margin of product introduction is more responsive to the drop in demand. However, we acknowledge that it is challenging to tightly associate these categories of new products to different types of innovation because carmakers may launch a new model name to refresh the image of a new generation of an existing model.

3.2.3 Geographical Variation: Ohio versus Texas

We now exploit geographical variation across states to connect the dynamics of the dispersion of expenditures with the depth of the recession. This decomposition isolates the role

Appendix B displays the time series of the number of transactions on new models N_t^N , as well as the share of models we classify as new.

of household demand for quality, because the set of products is constant across states. In particular, we observe variation in the magnitude of the drop in the dispersion of expenditures and a larger quality adjustment in states where households were hit more strongly by the recession.

To document this pattern, we compare Ohio and Texas, for two main reasons. First, they are the two largest states in our data and account for the bulk of transactions (approximately 80%). Second, Ohio and Texas experienced starkly different macroeconomic dynamics during the Great Recession, making the comparison of these two states insightful.

To highlight the macroeconomic differences between Ohio and Texas, we follow the approach of [Gertler and Gilchrist \(2018a\)](#), which analyzes state-level variation in the intensity of the Great Recession, focusing on house-price and employment dynamics (see also the related approach of [Mian, Rao, and Sufi, 2013](#)). The top panels of Figure 3 portray the Federal Housing Authority house-price index (top-left) and total non-farm employment (top-right) in Ohio (solid lines) and Texas (dashed lines). Ohio experienced a deep recession, with a 10% home price decline and an 8% employment decline. In contrast, Texas did not experience any housing bust and its decline in employment was less significant.

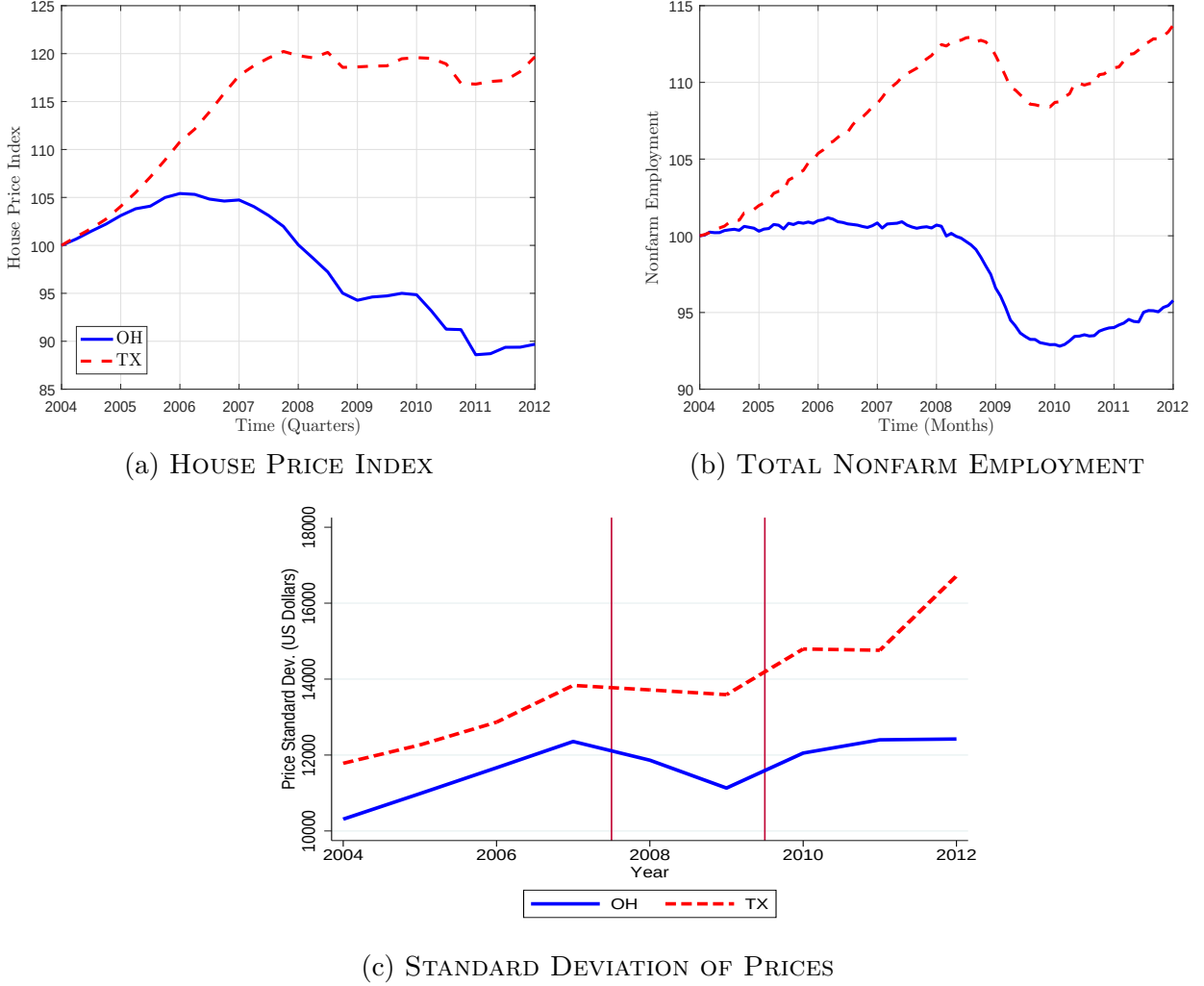
Geographical heterogeneity in the depth of the recession is likely associated with variation in household *demand* for durable-goods quality. Accordingly, Ohio experienced a downward adjustment in the demand for quality more sizable than Texas: The bottom panel of Figure 3 displays the standard deviation of the distribution of transaction prices in these two states and shows that the dispersion in Ohio (solid line) dropped more significantly than in Texas (dashed line) during the Great Recession. Consistent with a differential drop in demand for quality, we also find that the compression in the distribution of expenditures in Ohio is primarily due to a relative decline in the median and in higher percentiles, whereas these changes are less pronounced in Texas.

3.3 Dynamics of the Distribution of Quality

Our decompositions establish that the heterogeneity between models and, critically, new models are the main drivers of the dynamics of the distribution of new-car expenditures. Moreover, quality differences between new and continuing models were lowest during the recession. These patterns spur us to study vehicle characteristics.

To this end, we use hedonic regressions to estimate the function that maps vehicle characteristics to prices (for a seminal contribution, see [Griliches, 1961](#)). Formally, let the

Figure 3: Ohio versus Texas



Notes: The figure displays the dynamics of house prices, employment, and dispersion of expenditures on new cars in Ohio and Texas around the Great Recession. The top-left panel displays the quarterly Purchase Only Index of house prices from the Federal Housing Authority and the top-right panel displays monthly Total Nonfarm Employment from the Bureau of Labor Statistics ([Gertler and Gilchrist, 2018b](#)). Both series are normalized to equal 100 in both states at the beginning of 2004. The bottom panel displays the standard deviation of the distribution of transaction prices from the Dominion dataset. Horizontal axes report years. Solid lines refer to Ohio, dashed lines to Texas.

average price p_{it} of car model i in year t equal:

$$p_{it} = h_t(X_{it}, W_{it}, \eta_{it}),$$

where $h_t(\cdot)$ is the hedonic function; X_{it} are observed continuous vehicle attributes, such as fuel efficiency, horsepower, engine size, weight, and wheelbase; W_{it} are observed discrete

attributes, such as indicator variables for make, four-wheel drive, number of gears, manual transmission, turbo injection, number of cylinders, diesel, number of seats, and number of doors; and η_{it} are unobserved determinants of prices. We transform all continuous variables in logarithms and assume that the log of the hedonic function $h_t(\cdot)$ is linear:

$$\log p_{it} = \beta_t \log X_{it} + \gamma_t W_{it} + \eta_{it}, \quad (1)$$

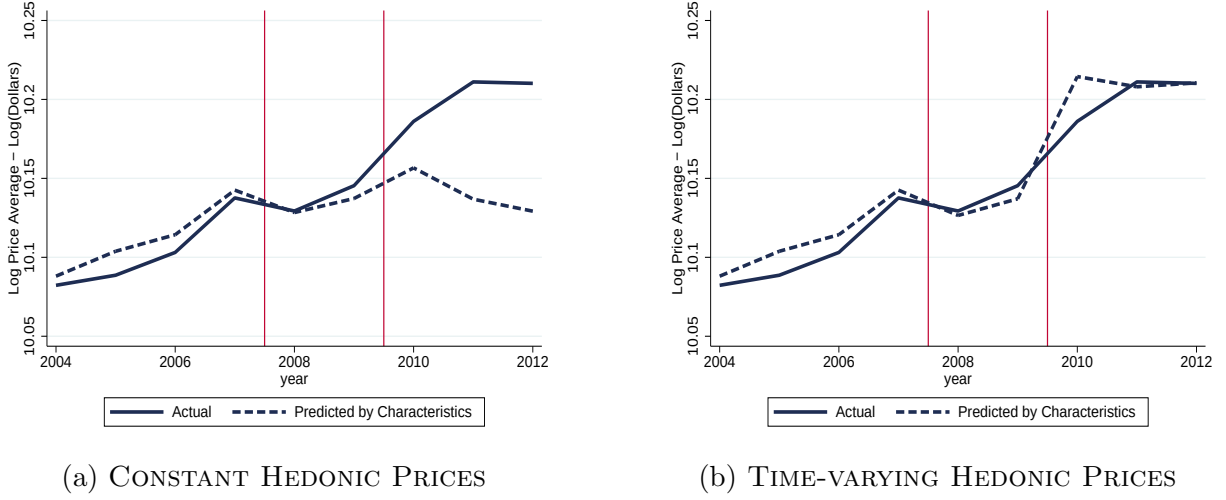
where β_t and γ_t are the vectors of coefficients, or “hedonic prices” of car characteristics.

We observe detailed characteristics of different trims of each model in the IHS dataset, whereas we observe transaction prices at a coarser level of aggregation—namely car models—in the merged dataset. Thus, we aggregate all continuous characteristics of different trims of each model, weighting different trims according to their transaction shares in the IHS dataset, whereas we consider different discrete characteristics as different observations, or, equivalently, different models.

We consider three subsamples: pre-recession (2004-2007), recession (2008-2009), and post-recession (2010-2012), assuming the coefficients are constant within each subsample but are potentially different across subsamples. We use these hedonic regressions to implement decompositions between the differences in the mean characteristics of vehicles over time and the differences in the hedonic prices of these characteristics over time (Oaxaca, 1973; Blinder, 1973). We leverage these estimates to track the evolution of the distribution of quality, by assigning a predicted value based on characteristics to each model. Formally, given the estimated hedonic prices $\hat{\beta}_{2004-2007}$ and $\hat{\gamma}_{2004-2007}$, we measure the quality of vehicle j in year $t = 2004, 2005, \dots, 2012$ as $\hat{\beta}_{2004-2007} \log X_{jt} + \hat{\gamma}_{2004-2007} W_{jt}$. This prediction represents the value of the bundle of characteristics contained in model j in year t , based on the dollar value of these characteristics implicit in pre-recession prices.

The left panel of Figure 4 displays our third main finding, which relates the dynamics of average price and average quality during and after the recession. The panel shows that they grow at a similar rate until the recession and, crucially, quality predicts the decline in the average price during the recession. In fact, the decline in average quality between 2007 and 2008 is slightly larger than the decline in the average price. We relate the dynamics of prices to the dynamics of selected characteristics during the recession analyzing the evolution of several variables associated with high quality based on our hedonic regressions, such as wheelbase, horsepower, and engine size. The averages of all these characteristics decline during the recession, which suggests a reallocation of expenditures toward smaller and less powerful cars, consistent with the dynamics of prices displayed in Figure 1.

Figure 4: Hedonics and Vehicle Quality



Notes: The figure displays the dynamics of average (log) transaction price in the merged Dominion-IHS dataset (solid lines) and the average (log) value predicted with a hedonic regression—equation (1)—(dashed lines). Each model is weighted according to its transaction share in the IHS dataset. The left panel refers to constant pre-recession hedonic prices (2004-2007); the right panel to time-varying hedonic prices, estimated in three subsamples: pre-recession (2004-2007), recession (2008-2009), and post-recession (2010-2012). Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

However, the left panel of Figure 4 displays a striking pattern from 2009 onward. Specifically, the average price grows at a rate of approximately 2% per year, whereas the average value implied by car characteristics declines protractedly, diverging from the average price until the end of our sample. Notably, average quality shows no growth in 2007-2012, while the average price grows by 7%.

This apparent decoupling between prices and predicted quality, based on pre-recession prices, indicates that the post-recession hedonic prices of some characteristics are higher than their pre-recession values. Different car attributes or brands may have different costs or may be valued differently over time, implying that changes in the state of the economy likely affect hedonic prices (Pakes, 2003). Accordingly, we re-estimate equation (1) separately in the three subsamples, and use these different estimates to compute a second measure of average quality. The right panel of Figure 4 displays the dynamics of this second measure of average quality, based on time-varying hedonic prices. The panel shows that this measure of average quality tracks the average price closely in all sub-periods.

The difference between our first and our second measures of quality confirms that the hedonic prices of some characteristics increased over time. Specifically, the hedonic prices of two important characteristics—wheelbase and horsepower—increased by over 20% in

the post-recession sample relative to the pre-recession sample. Changes in the hedonic prices of characteristics associated with high quality have different potential explanations, including a relative scarcity of models in the most expensive segments or time-varying markups. Nevertheless, this increase in the price of quality may partially account for the slow recovery in new-car sales after the Great Recession.

Critically, we find that pre-recession hedonic prices accurately predict the dynamics of expenditures on new models during the recession. The hedonic regression accounts for approximately 98% of the observed drop in between-model dispersion of new-model prices, though it slightly overpredicts the decrease in their average price. These results confirm that reallocation across different levels of quality accounts for the dynamics of the distributions of expenditures on all and new models.

In Appendix B, we analyze geographical heterogeneity in the dynamics of car quality, estimating separate hedonic regressions in Ohio and Texas. Both states experience a decline in average prices relative to their respective trends. However, in Ohio—where the recession was deeper—we observe a larger substitution toward models with lower quality, as well as a larger and more persistent gap between price and quality, which buttresses the primary role of household demand for quality.

Finally, we investigate any differential effects between US and foreign carmakers. While the hedonic regressions show that the point estimates of US carmaker fixed effects are lower than those of Asian and European carmakers, respectively, the estimates do not show differential changes across periods.

Overall, our hedonic regression analysis highlights some striking dynamics in the quality of vehicles and confirms a reallocation in expenditures away from high-quality new models. In the next subsection we present a complementary analysis that focuses on technological trade-offs in the set of models available on the market, abstracting from information on prices. This analysis allows us to address some potential limitations of the hedonic methodology, such as the difficulty of disentangling changes in marginal costs from changes in markups and in preferences for different models that may occur around the recession.

3.4 New Models and Technological Progress

We now analyze the level of technology embodied in vehicles and document a sharp drop in the quality of new models introduced during the Great Recession. This analysis allows us to isolate the role of supply factors for the downward quality adjustment in durable goods.

We follow [Knittel \(2011\)](#) to measure the technological trade-off between fuel efficiency,

weight, and engine power, and to estimate its evolution over time. This methodology posits a marginal-cost function that depends on vehicle attributes and estimates the level sets of this function, using time fixed effects to capture the evolution of the technological frontier. Specifically, the marginal cost function for vehicle i in year t equals:

$$c_{it} = c_t^1(\text{mpg}_{it}, \text{hp}_{it}, w_{it}, Z_{it}^1, \mathcal{I}_{it}^N) + c_t^2(Z_{it}^2),$$

where $c_t^1(\cdot)$ is the component of marginal cost related to fuel economy, which depends on fuel efficiency mpg_{it} , horsepower hp_{it} , weight w_{it} , a subset of characteristics Z_{it}^1 that are relevant for the trade-off of interest, and $\mathcal{I}_{it}^N$ is an indicator variable for new models; $c_t^2(\cdot)$ is the component of the marginal cost that depends on other characteristics that are less related to fuel economy, Z_{it}^2 . We include a large set of indicator variables for vehicle characteristics Z_{it}^1 , such as make, diesel engine, turbo injection, manual transmission (also interacted with a time trend).

We further assume that vehicle attributes enter the marginal-cost function $c_t^1(\cdot)$ in a log-linear form—i.e., the cost function is Cobb-Douglas—and that time t affects this function in multiplicative form—i.e., technological progress is input neutral. Under these assumptions, we estimate the level sets of the marginal cost $c_t^1(\cdot)$ with the following specification:

$$\log \text{mpg}_{it} = \alpha_{hp} \log \text{hp}_{it} + \alpha_w \log w_{it} + \alpha_Z Z_{it}^1 + \alpha_N \mathcal{I}_{it}^N + T_t + T_t \times \mathcal{I}_{it}^N + \varepsilon_{it}, \quad (2)$$

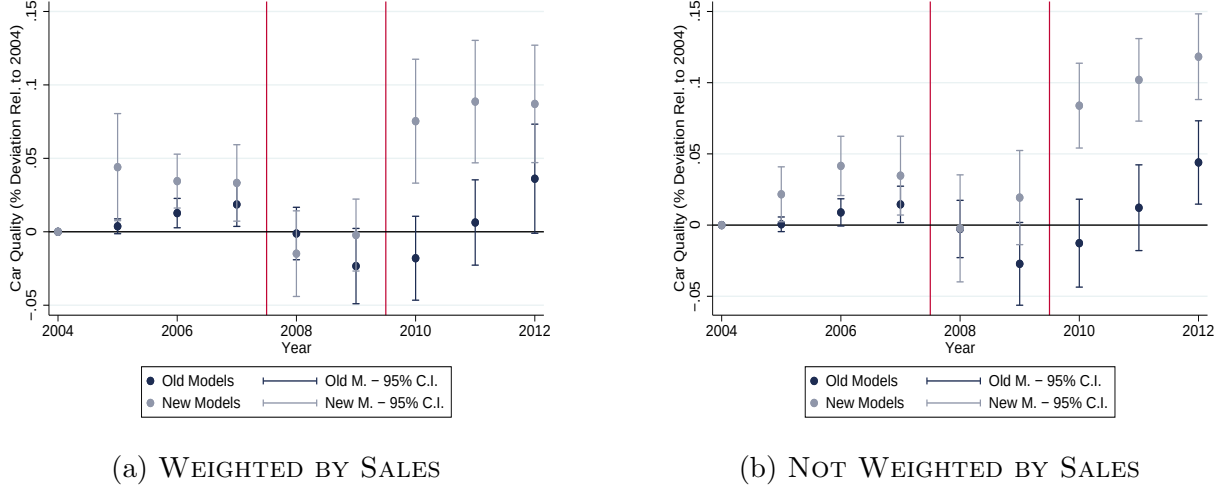
where T_t is a year fixed effect; $T_t \times \mathcal{I}_{it}^N$ is the interaction between time fixed effects and the indicator variable for new models, which allows the regression (2) to flexibly capture a differential effect of the recession on new models; and ε_{it} are unobservables.

Whereas the hedonic approach combines the reallocation of demand and changes in the supply of quality, the marginal-cost estimation likely highlights quality changes that originate on the supply side of the market. We estimate equation (2) in two ways, first weighting models by the number of transactions, and then without sales weights, which further isolates changes in the quality of products supplied.¹¹

Figure 5 displays the estimated year fixed effects for new models (clear markers) and old models (dark markers), relative to their pooled baseline value in 2004, normalized to zero. The left panel portrays the estimates of the sales-weighted regression and the right panel refers to the unweighted regression. In both cases, we find that, typically, the level

¹¹In both cases, for consistency with our analysis of Section 3.3, we aggregate all continuous characteristics of different trims of each model at the model level, weighting different trims according to their transaction shares, and we consider different discrete characteristics as different observations.

Figure 5: Technology of New and Old Models



Notes: The figure displays the estimated average level of technological efficiency for new models (clear markers) and old models (dark markers), measured as the estimated time fixed effects in regression equation (2). The left panel refers to a regression with weights based on the number of transactions in the IHS dataset, whereas the right panel refers to a regression without weights. The horizontal axis reports years (2004-2012); vertical lines highlight recession years (2008 and 2009).

of technology grows over time, with new models displaying superior technology than old models. However, our fourth main finding is that, during the Great Recession, the growth rate of quality of new models declines: In 2008, the estimated quality of new models is similar to the quality of old models, which implies a halt in the adoption of frontier technologies embodied in new vehicles. Consistent with this drop in technology adoption, we estimate that the quality of continuing models also declines in 2008 due to an inferior mix of characteristics.

Quantitatively, the coefficients displayed in Figure 5 mean that the average level of technology of new models declines by almost 5% between 2006 and 2008. The similarity of the left and right panels supports the notion that the main driver of this decline is that the quality of newly introduced vehicles drops in the recession.

We further estimate the technology levels separately for models introduced by European, Asian, and US carmakers. We find that the drop in new-product quality is largest for European carmakers, which on average specialize in high-quality models. This finding, along with our finding on the crucial role of European carmakers for high-quality models (Section 3.2.2), supports our interpretation that the downward quality adjustment on the supply side is likely an endogenous response to the drop in demand, and less likely due to

other shocks hitting carmakers, such as financial shocks, which were more severe for US manufacturers.¹²

Although the technological level of new models recovers sharply from 2010, the low quality of new models introduced during the recession—which remain in the set of available models for several years—persistently drags the average level of technology for the continuing models, which remains on a lower path throughout the recovery. Overall, the technological level of old models breaks its pre-recession 2007 level only at the end of our sample, as models introduced during the recession are gradually replaced.

Accordingly, we perform a back-of-the-envelope calculation of the effects of these dynamics on the average quality of the overall stock of registered cars, combining our estimated level of technology for new cars with information on new-car registrations during the period of our analysis. Appendix B.4 provides the details of this calculation. We estimate that by 2012, the quality of the car stock was 1.3% lower than if new-car technology and new-car registrations had remained on their pre-recession trends. The drop in new-car quality accounts for almost one percentage point of this decline.

4 Concluding Remarks

Our analysis shows that both demand and supply factors contributed to a downward quality adjustment in expenditures during the Great Recession. Amid a decline in demand and a reallocation of expenditures away from expensive models, automakers introduced models of low quality, leading to a persistent decline in technology.

We argue that alternative mechanisms that affect exclusively demand or supply cannot fully account for all patterns on quality dynamics that we uncover. Geographical variation in expenditures highlights the critical role of household demand for quality, thus inconsistent with supply shocks hitting only manufacturers—such as financial shocks—determining quality dynamics. Our finding that manufacturers modified the path of technology embodied in new models points to an important role for supply, inconsistent with an explanation based exclusively on household demand, through substitution toward lower quality within a fixed set of products.

We believe that this evidence will prove useful in informing quantitative models of innovation over the business cycle.

¹²Furthermore, we find that the technology level evolves similarly for new model names and new generations of existing model names.

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APPENDICES

A Data and Measurement

In this appendix, we describe our procedure for merging the Dominion dataset and the IHS dataset and explain our definitions of vehicle models.

A.1 Merging Dominion and IHS Datasets

For each transaction in the Dominion dataset, we observe a string for make name—e.g., “TOYT” for Toyota—and a string for model name—e.g., “Camry”—as well as the corresponding model-year, which may or may not correspond with the calendar year in which the transaction takes place because new models marketed as model-year t are often introduced in year $t - 1$.

For each vehicle model in the IHS dataset, we observe a string for make name—e.g., “Toyota”—and a string for model name—e.g., “Toyota Camry”—as well as a variable named generation-year, which allows us to identify different generations of a same model—e.g., first generation, second generation, etc. Moreover, we also observe the total number of US transactions by calendar year.

1. In the Dominion dataset we identify all strings corresponding to make and model names.
2. We perform the same step, identifying make and model names in the IHS dataset.
3. For all make-model names in the Dominion dataset (point 1), we find a single corresponding make-model name in the IHS dataset (point 2). Whenever we do not find a match for the make-model name (approximately 19% of cases), we assign as model name the combination of make name and the first word of the model string from the Dominion dataset.
4. For each make-model name in the Dominion dataset, we identify the corresponding set of model-years for which we observe a positive number of transactions. For example, in the case of the Toyota Camry, these model-years are 2003, 2004, ..., 2013.
5. For each make-model-generation in the IHS dataset, we identify the first model-year with a positive number of transactions in the IHS dataset. If the first year with a positive number of transactions of a make-model-generation is year t , we infer that

the first model-year for that make-model-generation is year $t + 1$, to account for the fact that vehicles marketed as model-year t are typically first introduced in the market in year $t - 1$.

6. We merge the dataset of Dominion make-model-years (point 4) with the Dominion-IHS matched list of make-model names (point 3).
7. We assign each make-model-year from the Dominion dataset (point 6) to the corresponding make-model-generation (point 5) as follows: Toyota Camry model-years 2007-2011 are assigned to the generation-year 2007 and Toyota Camry model-years 2012 through 2013 are assigned to generation-year 2012.

A.2 Model Definitions

We define a vehicle model as a triplet of make, model, and generation obtained following the merging procedure described above—e.g., Toyota Camry generation-year 2007.

We define a new model in year t as a model for which we observe the first transaction in year t or in year $t - 1$, to account for the fact that the first transaction on a new model tends to appear in the second half of the year. Specifically, this implies that we consider a model as new whenever its model year in the Dominion dataset corresponds with its generation year, and possibly also whenever we observe a transaction for this model that occurs in a calendar year preceding its model year. Thus, this definition includes new model names as the first generation of a model, as well as new generations of existing model names. We exclude 2004 from our analysis of new models because this is the first year in the Dominion dataset, and thus we cannot cleanly identify new models.

We should point out that because we observe transaction prices at the model level in the Dominion dataset and, thus, we merge information from the Dominion dataset and the IHS dataset at the model level, there remains some residual heterogeneity in vehicle characteristics across different trims of each model in the IHS dataset. To deal with this heterogeneity, in our analyses of car characteristics in Sections 3.3 and 3.4, we average all continuous car characteristics across different trims of each model using their respective transaction shares in the IHS dataset, whereas we treat vehicles with different values of discrete characteristics—such as diesel, or turbo injection—as different models. In Appendix B we consider an alternative approach, aggregating both continuous and discrete characteristics at the model level using their transaction shares. As Figures B12 and B13 show, our main findings are robust to this alternative approach, suggesting that the level

of aggregation of car characteristics, as well as the exact number of models, do not affect our results.

B Additional Empirical Evidence

In this appendix, we provide additional empirical evidence and document several robustness checks.

B.1 Dynamics of the Distribution of Expenditures

Figure B1 displays aggregate consumer expenditures on durable goods (left panel) and on vehicles (right panel) during 2004-2012 (US Bureau of Economic Analysis, 2022) and shows the large drop in these components of household expenditures during the Great Recession.

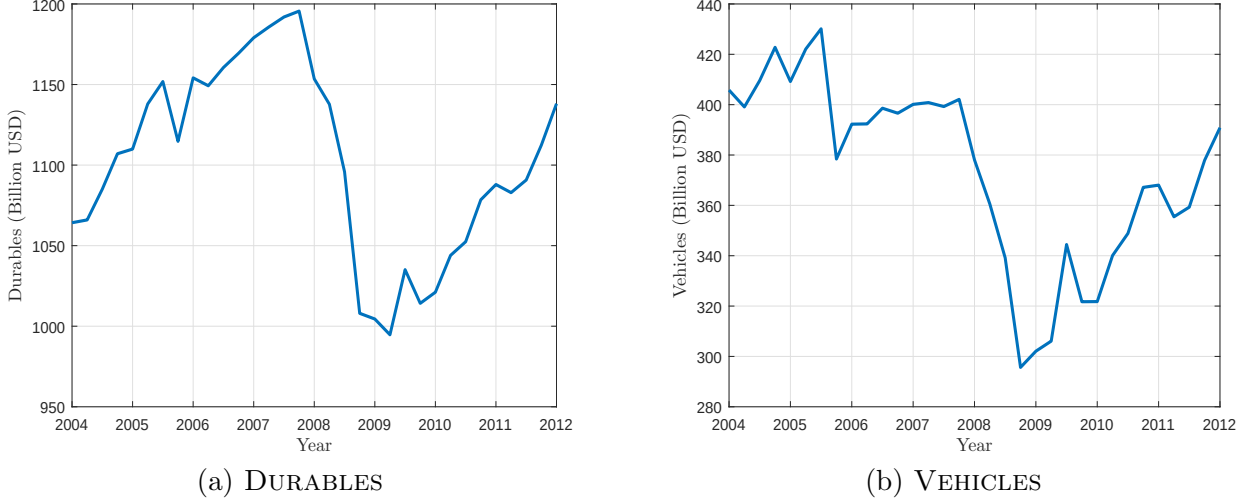
During July and August of 2009, the Car Allowance Rebate System, commonly known as “Cash for clunkers,” subsidized the replacement of highly polluting cars with new ones, potentially affecting the pool of new-car buyers (Hoekstra, Puller, and West, 2017). Figure B2 reproduces the findings displayed in Figure 1, but excluding the months of July and August in each year to show that the patterns of the distribution of expenditures on new vehicles are not significantly affected by the Cars Allowance Rebate System. Figure B3 displays the same variables, but excludes fleet sales—which account for approximately 4.4% of transactions—to show that our main findings are unchanged if we restrict attention to consumer sales only.

B.2 Decomposing the Dispersion of Expenditures

Figures B4 and B5 reproduce the findings displayed in Figure 2 under the same two robustness checks described above: namely, removing July and August to exclude the effects of “Cash for clunkers” and removing fleet sales, respectively.

Figure B6 portrays the path of the average transaction price for ten popular models. Specifically, we select the five models with the highest sales volume with price below the overall sample median, and the five models with highest sales volume with price above the median. For all of these models, the figure shows that prices did not significantly deviate from trend during the Great Recession. This confirms that reallocation between models, instead of price changes at the model level, account for changes in the distribution of expenditures during the recession. Consistent with this evidence, Gavazza and Lanteri

Figure B1: Consumer Expenditures on Durable Goods and on Motor Vehicles



Notes: The figure displays personal consumption expenditures on durable goods (left panel) and on motor vehicles and parts (right panel), at quarterly frequency, seasonally adjusted annual rate, from the Bureau of Economic Analysis during 2004-2012.

(2021) show that price changes during the Great Recession were concentrated in used-car markets.

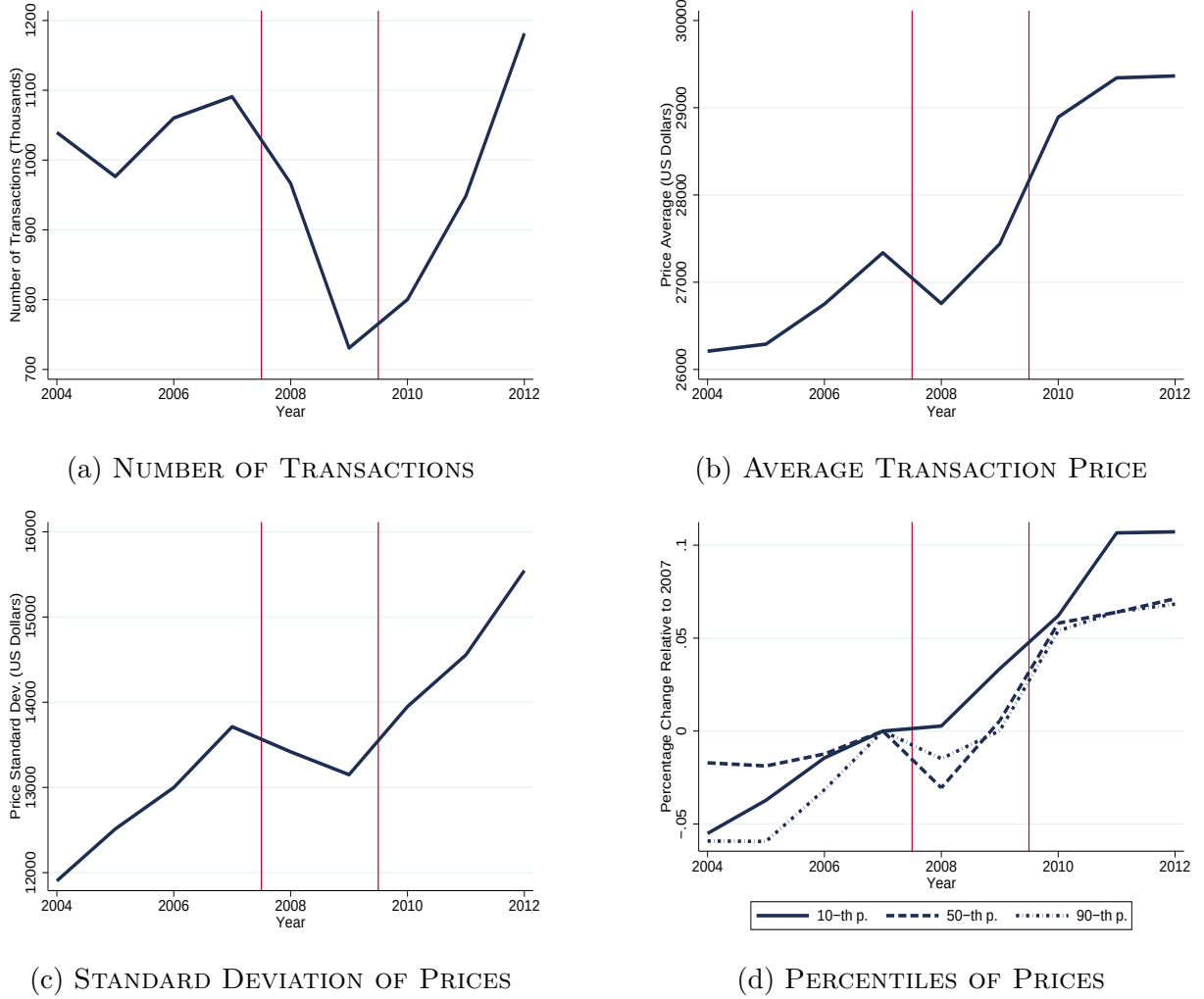
We further verify that a reallocation of market shares across different models accounts entirely for our findings on the dynamics of between-model price variance—as well as average price—by performing the following analysis. We decompose the difference between the between-model variance in year t , V_t^B , and the between-model variance in the whole sample, V^B , as follows:

$$\begin{aligned}
 V_t^B - V^B &= \sum_{i \in M_t} s_{it} (\bar{p}_{it} - \bar{p}_t)^2 - \sum \bar{s}_i (\bar{p}_i - \bar{p})^2 \\
 &= \sum_{i \in M_t} (s_{it} - \bar{s}_i) (\bar{p}_i - \bar{p})^2 + \sum_{i \in M_t} \bar{s}_i [(\bar{p}_{it} - \bar{p}_t)^2 - (\bar{p}_i - \bar{p})^2] + \\
 &\quad + \sum_{i \in M_t} (s_{it} - \bar{s}_i) [(\bar{p}_{it} - \bar{p}_t)^2 - (\bar{p}_i - \bar{p})^2],
 \end{aligned} \tag{B1}$$

where the index i refers to models, t denotes years, and we use bars to denote averages of prices p_{it} and market shares s_{it} .

The first term on the second line of equation (B1) measures the role of reallocation of expenditures across models; the second term on the second line measures the role of changes in model prices; and the final term denotes the covariance term between changes

Figure B2: Dynamics of New-Vehicle Expenditures, Excluding July and August of Each Year



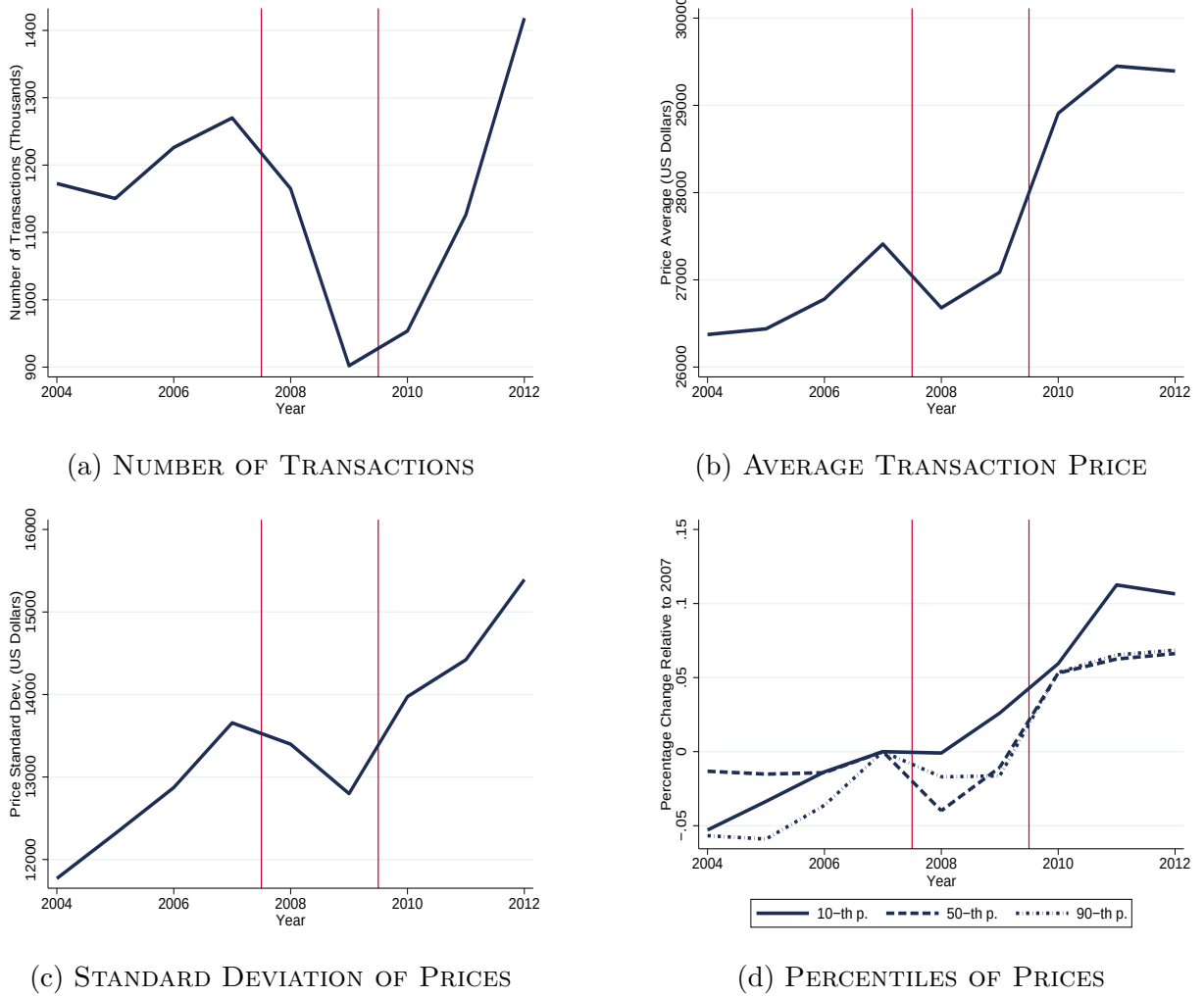
Notes: The figure displays the number of new-car sales (top-left panel), the average (top-right panel), the standard deviation (bottom-left panel), and three percentiles—10th, 50th, and 90th—(bottom-right panel) of the distribution of transaction prices from the Dominion dataset, excluding the months of July and August of each year. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

in market shares and prices.

Figure B7 shows that between 2007 and 2009, the component due to reallocation (dashed line) accounts for the entire decline in between-model variance. This finding buttresses our interpretation of the variance decompositions in Section 3.2: A reallocation of expenditures toward models of lower quality accounts for the drop in dispersion of expenditures.

We also perform the same decomposition for the average price in year t (equation (B2))

Figure B3: Dynamics of New-Vehicle Expenditures, Excluding Fleet Sales



Notes: The figure displays the number of new-car sales (top-left panel), the average (top-right panel), the standard deviation (bottom-left panel), and three percentiles—10th, 50th, and 90th—(bottom-right panel) of the distribution of transaction prices from the Dominion dataset, excluding fleet sales. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

and obtain again a tight match between the overall average-price dynamics around the recession and the component due to reallocation of market shares across models, as Figure B8 shows.

$$\begin{aligned}
 \bar{p}_t - \bar{p} &= \sum_{i \in M_t} s_{it} \bar{p}_{it} - \sum \bar{s}_i \bar{p}_i \\
 &= \sum_{i \in M_t} (s_{it} - \bar{s}_i) \bar{p}_i + \sum_{i \in M_t} \bar{s}_i (\bar{p}_{it} - \bar{p}_i) + \sum_{i \in M_t} (s_{it} - \bar{s}_i) (\bar{p}_{it} - \bar{p}_i).
 \end{aligned} \tag{B2}$$

Furthermore, we obtain similar results when we restrict attention to the variance of prices of new models.

Figure B9 displays the time series of the total number of sales and the number of sales of new models (left panel) and the share of models we classify as new models (right panel). These two figures show that both the share of transactions on new models and the flow of new-product introduction are procyclical, peaking in 2007 and dropping during the Great Recession.

Figure B10 displays our findings on the patterns of new-model introduction across carmakers of different geographical origin (Europe, Asia, and US). The left panel shows that the number of new models dropped for all three groups during the recession. Between 2007 and 2009, the volume of new-model introduction dropped by for European carmakers, by for Asian carmakers, and by for US carmakers. As a result, there is a missing generation of new models across all makes. After the recession, we observe some heterogeneity in the speed of recovery, with European carmakers increasing new-model introduction faster than Asian and US carmakers.

The right panel of Figure B10 focuses on new models with an average price above \$40,000, which approximately corresponds to the 90th percentile of the distribution in 2007. In this range, we observe that European carmakers account for the majority of new models. In 2007, out of 24 new models introduced by European carmakers, 17 are above the \$40,000 price threshold. In contrast, out of 26 new models introduced by US carmakers, only 5 are above the same threshold; the fraction of high-price new models introduced by Asian carmakers is even smaller. As the right panel of Figure B10 shows, high-price new-model introduction from European carmakers dropped almost by half during the recession, which largely accounts for the missing generation of high-quality new models in 2008 and 2009.

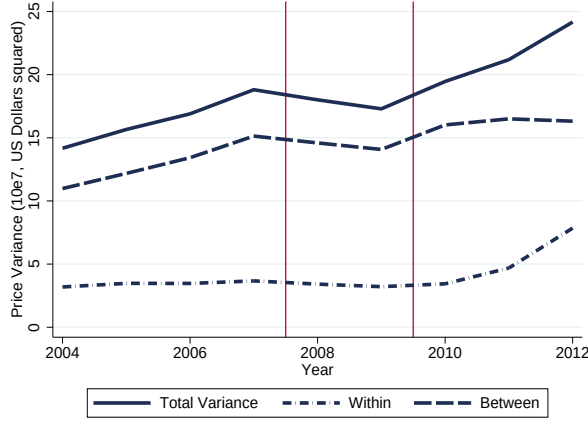
Figure B11 displays the decomposition of new-model introduction into new model names and new generations of existing model names.

B.3 Dynamics of the Distribution of Quality

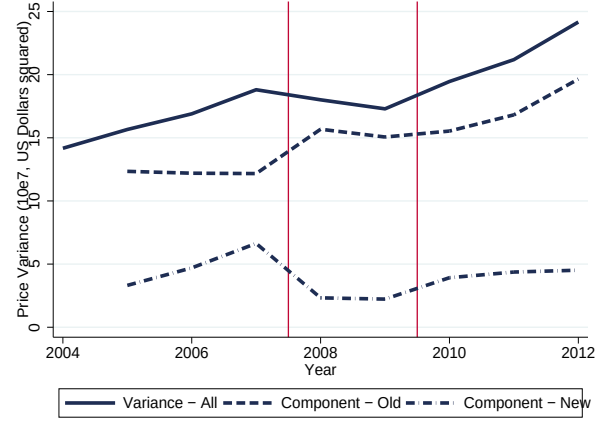
Table B1 reports the results of our hedonic-regression analysis. Specifically, column (1) in Panel A of Table B1 reports the hedonic prices of the main continuous attributes X_{it} in the pre-recession subsample. Columns (2) and (3) in Panel A report the hedonic prices of the main continuous attributes X_{it} in the recession and post-recession subsamples, respectively.

These hedonic regressions are well suited for accounting for the dispersion of expendi-

Figure B4: Variance Decomposition, Excluding July and August of Each Year



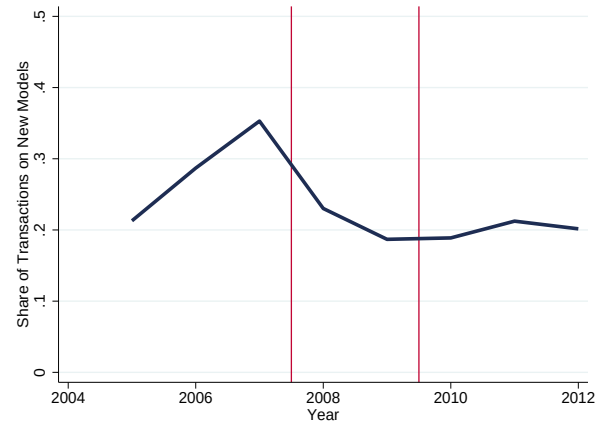
(a) VARIANCE OF NEW-CAR PRICES, BETWEEN AND WITHIN



(b) VARIANCE OF NEW-CAR PRICES, NEW AND CONTINUING



(c) VARIANCE OF NEWLY INTRODUCED MODELS

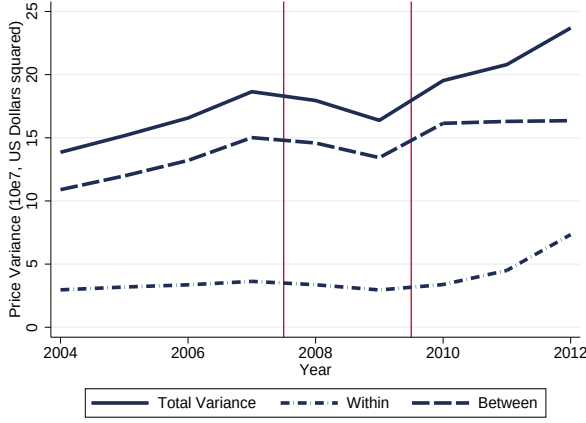


(d) SHARE OF NEWLY INTRODUCED MODELS

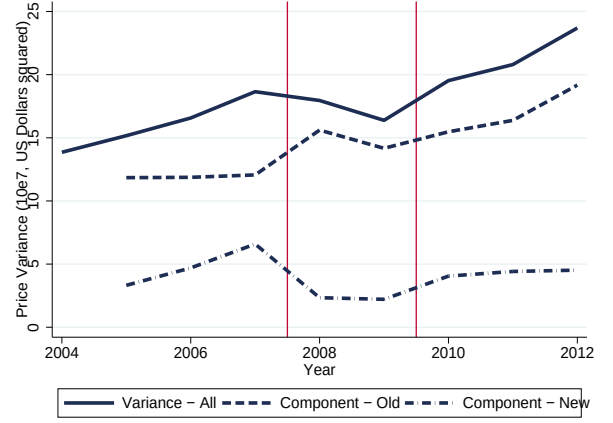
Notes: The figure displays several decompositions of the variance of transaction prices in the Dominion dataset, excluding the months of July and August of each year. The top-left panel displays the decomposition of the variance of new-vehicle transaction prices V_t (solid line) into the following components: between models V_t^B (dashed line) and within models V_t^W (dashed-dotted line). The top-right panel displays the decomposition of the variance V_t (solid line) into two components: new models $s_t^N V_t^N$ (dashed-dotted line) and old models $(1 - s_t^N) V_t^O$ (dashed line). The bottom-left panel displays the variance of expenditures on new models V_t^N (solid line) and its decomposition into between-models component $V_t^{N,B}$ (dashed line) and within-models component $V_t^{N,W}$ (dashed-dotted line). The bottom-right panel displays the share of transactions on new models s_t^N . Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

tures. Car characteristics capture a large share of the between-model variance in prices: R^2 coefficients of the hedonic regressions exceed 0.93 in all subsamples.

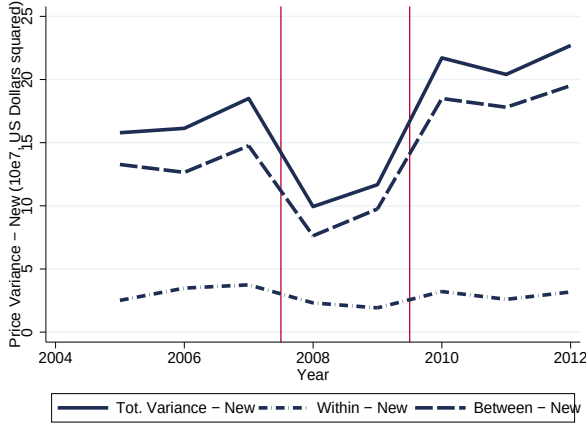
Figure B5: Variance Decomposition, Removing Fleet Sales



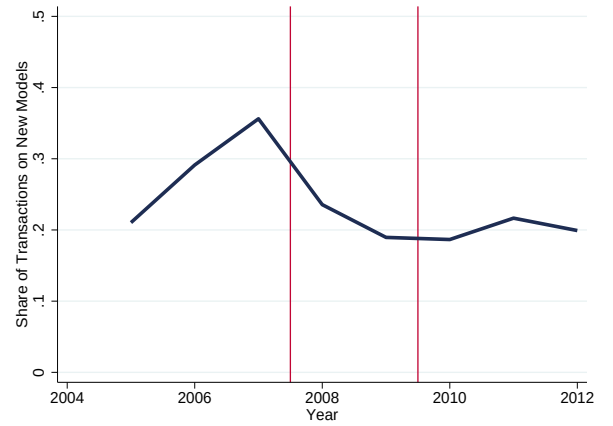
(a) VARIANCE OF NEW-CAR PRICES, BETWEEN AND WITHIN



(b) VARIANCE OF NEW-CAR PRICES, NEW AND CONTINUING



(c) VARIANCE OF NEWLY INTRODUCED MODELS

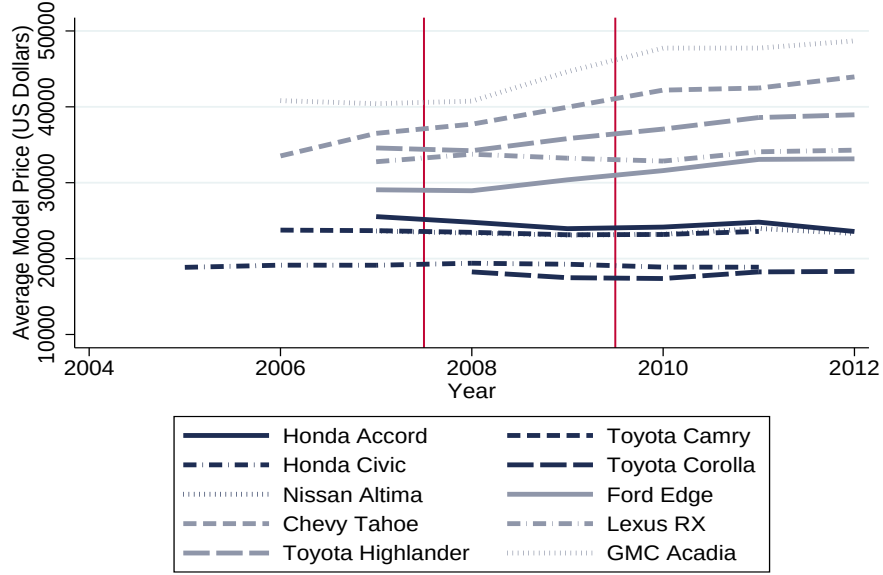


(d) SHARE OF NEWLY INTRODUCED MODELS

Notes: The figure displays several decompositions of the variance of transaction prices in the Dominion dataset, excluding fleet sales. The top-left panel displays the decomposition of the variance of new-vehicle transaction prices V_t (solid line) into the following components: between models V_t^B (dashed line) and within models V_t^W (dashed-dotted line). The top-right panel displays the decomposition of the variance V_t (solid line) into two components: new models $s_t^N V_t^N$ (dashed-dotted line) and old models $(1 - s_t^N) V_t^O$ (dashed line). The bottom-left panel displays the variance of expenditures on new models V_t^N (solid line) and its decomposition into between-models component $V_t^{N,B}$ (dashed line) and within-models component $V_t^{N,W}$ (dashed-dotted line). The bottom-right panel displays the share of transactions on new models s_t^N . Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

The table shows that the coefficients of some attributes, most notably engine size, are not precisely estimated. The reason is that our regression equation (1) includes some discrete characteristics W_{it} , such as indicator variables for the number of cylinders, which

Figure B6: Average Price of Ten Popular Models



Notes: The figure displays the average transaction price of ten popular models in the Dominion dataset. Specifically, we select the five models with the highest levels of sales and price below the median, and the five models with the highest levels of sales and price above the median. Horizontal axes report years (2004-2012); vertical lines highlight the recession years (2008 and 2009).

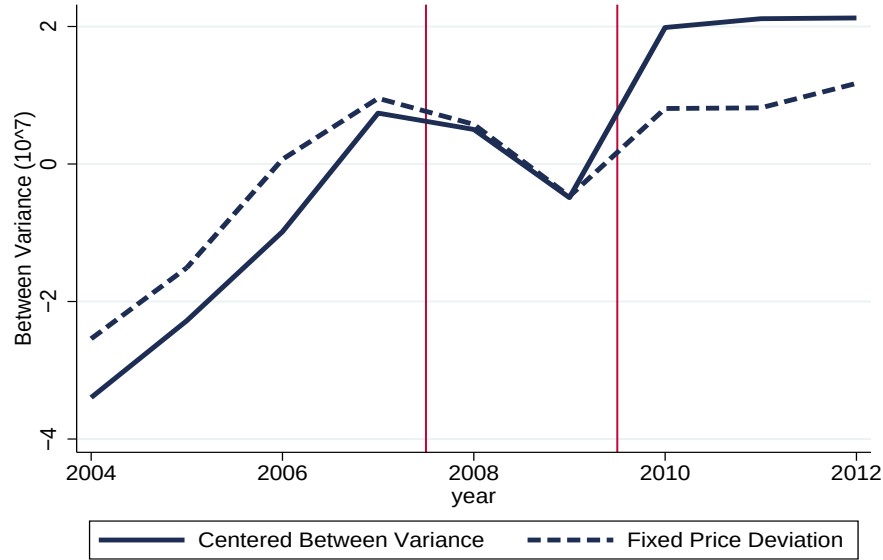
absorb almost all variation in engine size. Hence, the residual variation in engine size is minimal and its coefficient estimate is noisy.

Panel B of the table reports the peak-to-trough dynamics of expenditures and quality of newly introduced models.

Figure B12 displays the results of robustness analyses of average quality dynamics measured with hedonic regressions. Specifically, while we produce Figure 4 in Section 3.3 by aggregating continuous characteristics of different trims at the model level, but considering trims with different discrete characteristics—such as diesel, or turbo injection—as distinct models, in these robustness analyses we aggregate both continuous and discrete characteristics of different trims of each model.

We consider two alternative specifications of the hedonic regressions. The first specification (top panels) is more flexible and uses indicator variables for discrete characteristics, as in equation (1). Within each model, we average the discrete characteristics weighting different trims according to their transaction shares. We then round the average to the closest discrete value, and set the corresponding indicator variable equal to one. The second specification (bottom panels) treats all characteristics that vary across trims—including discrete ones—as continuous variables and assumes a log-linear relationship between prices

Figure B7: Decomposition of Between-Model Variance: Role of Expenditure Reallocation



Notes: The figure displays the results of the decomposition of the between-model variance defined in equation (B1). The solid line refers to the overall between-model variance, whereas the dashed line refers to the component due to reallocation of expenditures across models, for fixed deviations of prices from their average. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

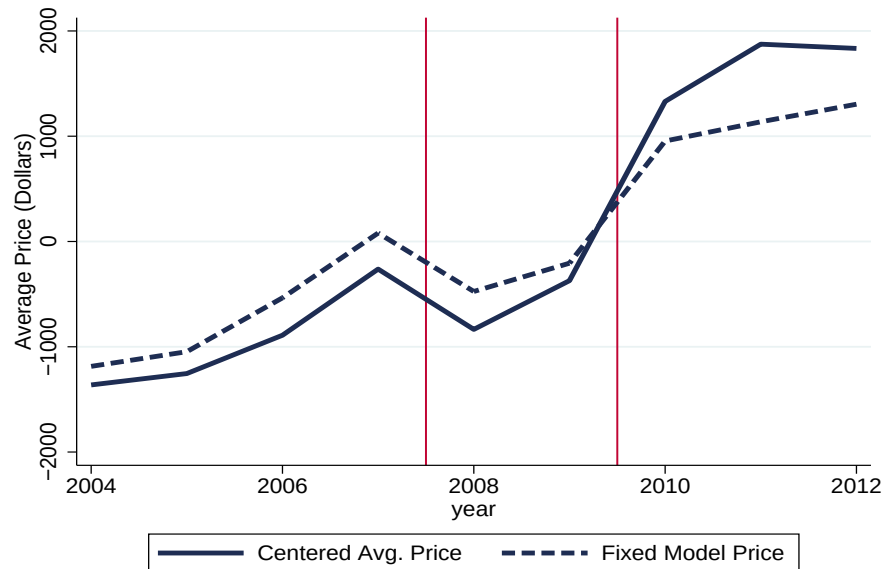
and all of these characteristics. Within each model, we average the discrete characteristics weighting different trims according to their transaction shares and treat the average as the value of a continuous characteristic. Because make and body type do not vary across trims within each model, we control for these two attributes with indicator variables as in equation (1).

The first specification has an overall better fit, because the indicator variables better capture the nonlinearities in the relation between discrete attributes—such as the number of cylinders—and prices, whereas the second specification features a finer measurement of discrete variables—as it does not rely on rounding—but imposes a linear relation between all attributes and prices.

Critically, in both cases we find that quality growth is stagnant after the Great Recession when we measure it with pre-recession hedonic prices (left panels), whereas average quality tracks the average price more closely when we use time-varying hedonic prices (right panels). These results suggest that the level of aggregation of car characteristics, as well as the exact number of models, do not affect our main findings.

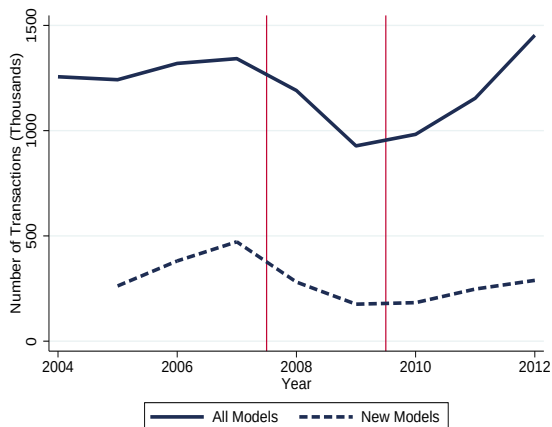
Table B2 reports selected coefficients of our hedonic regressions, with the same level of

Figure B8: Decomposition of Average Price: Role of Expenditure Reallocation

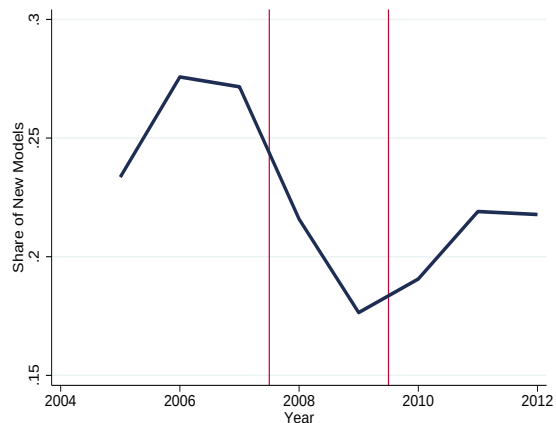


Notes: The figure displays the results of the decomposition of the average transaction price defined in equation (B2). The solid line refers to the overall average price, whereas the dashed line refers to the component due to reallocation of expenditures across models, for fixed average prices at the model level. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

Figure B9: Transactions and Share of New Models



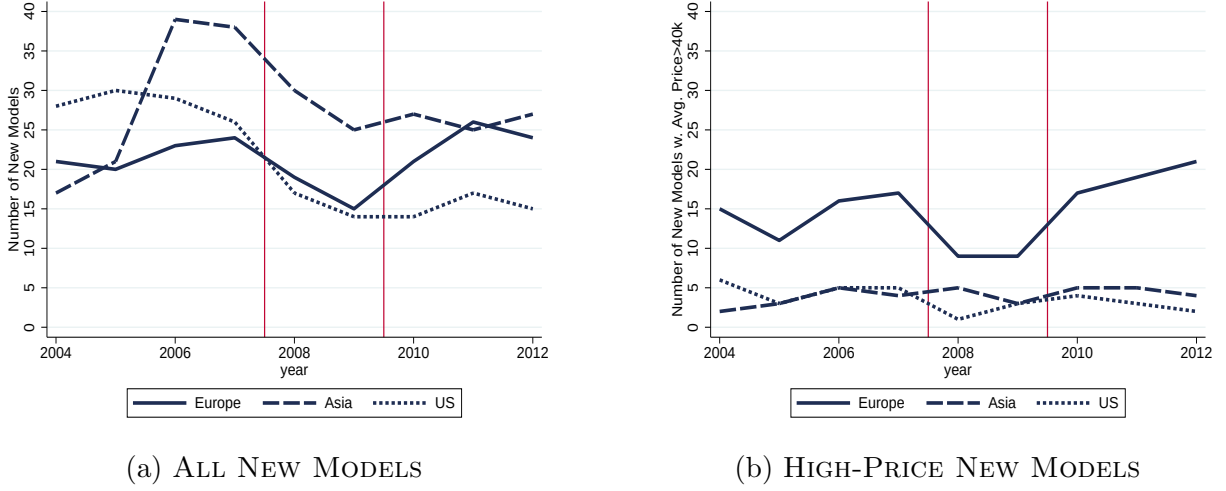
(a) NUMBER OF TRANSACTIONS



(b) SHARE OF NEW MODELS

Notes: The left panel displays the number and compositions of new-car sales in the Dominion dataset during 2004-2012. The solid line refers to all sales; the dashed line refers to sales of new-car models only. The right panel displays the time series of the share of models we classify as new models. Horizontal axes report years (2004-2012); vertical lines highlight the recession years (2008 and 2009).

Figure B10: Introduction of New Models by Origin of Carmakers



Notes: The figure displays the volume of new-model introduction by origin of carmakers. The left panel refers to all new models, whereas the right panel refers to new models with a price above \$40,000. Solid lines denote European carmakers; dashed lines denote US carmakers; and dotted lines denote Asian carmakers. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

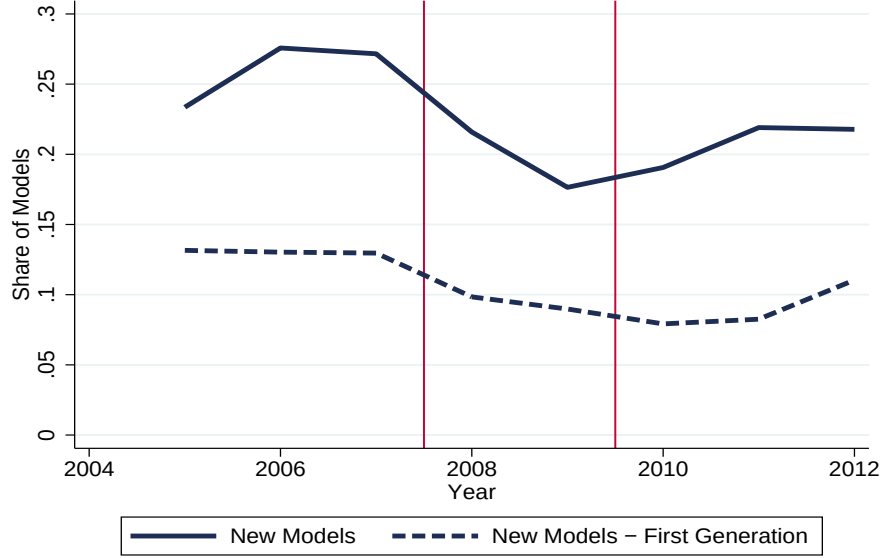
aggregation as in Section 3.3, when we focus exclusively on new models. Consistent with our baseline specification that pools all models (top panel of Table B1), we measure an increase in several hedonic prices of characteristics associated with high quality between the pre-recession and the post-recession periods.

We also explore geographical heterogeneity in the dynamics of the distribution of car quality. To this end, we reproduce the hedonic-regression analysis separately for transactions in Ohio and Texas. We first estimate the hedonic prices of car characteristics in the pre-recession period for each state. We then use these hedonic prices to measure the quality of all cars sold during and after the recession. Both states experience a decline in average prices relative to their respective trends. However, our estimates reveal that in Ohio the substitution toward lower-quality models during the recession is stronger than in Texas: The peak-to-trough decline in average quality is approximately equal to 2% in Ohio and 1.4% in Texas. Moreover, starting during the recession, Ohio features a larger gap between price and quality than Texas.

B.4 New Models and Technological Progress

Figure B13 reports several robustness checks of our estimates of the technology level for new and old models. Specifically, the top-left panel reports the results we obtain by replacing

Figure B11: Introduction of New Model Names



Notes: The figure displays a decomposition of the volume of new-model introduction between new model names and new generations of existing model names. The solid line refers to the total share of models that we classify as new and the dashed line refers to the share of models with a new model name in the Dominion dataset. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

the variable weight with three geometric dimensions—wheelbase, width, and height—in regression equation (2). Estimates of the technology level for new and old models are remarkably similar to the ones we show in Figure 2. Different from Knittel (2011), our dataset does not contain information about torque; thus, we measure engine power with horsepower across all of the specifications.

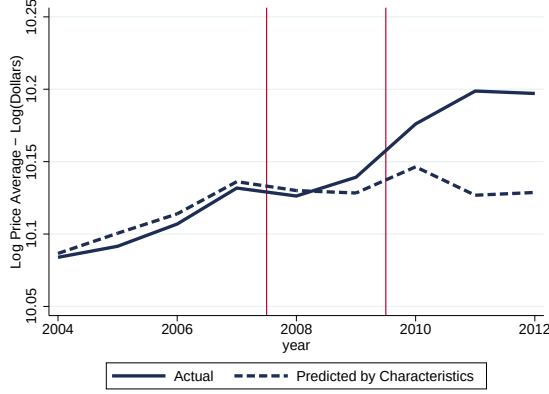
The top-right panel of Figure B13 displays our estimates of the technological level of new models and old models under the assumption of a translog cost function. Under this assumption, we recover the path of technological progress by estimating the following regression equation:

$$\begin{aligned} \log mpg_{it} = & \alpha'_{hp} \log hp_{it} + \alpha'_w \log w_{it} + \alpha'_Z Z_{it}^1 + \alpha'_N \mathcal{I}_{it}^N + T_t + T_t \times \mathcal{I}_{it}^N + \\ & \alpha'_{hp^2} (\log hp_{it})^2 + \alpha'_{w^2} (\log w_{it})^2 + \alpha'_{hp,w} \log hp_{it} \times \log w_{it} + \varepsilon_{it}. \end{aligned} \quad (B3)$$

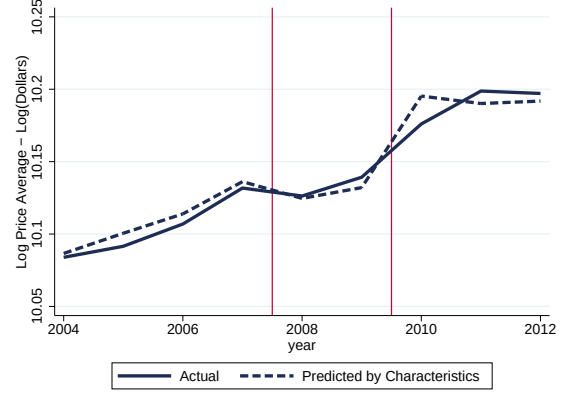
The results are qualitatively and quantitatively similar to those we obtain in Figure 5 under the assumption of a Cobb-Douglas cost function.

The bottom panel of Figure B13 displays our estimates of the technological level of new

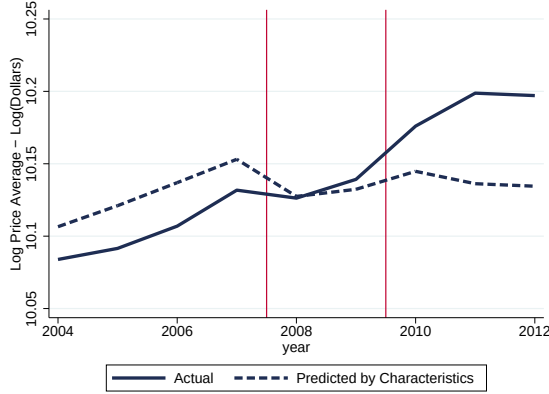
Figure B12: Hedonics and Vehicle Quality, Aggregating All Characteristics



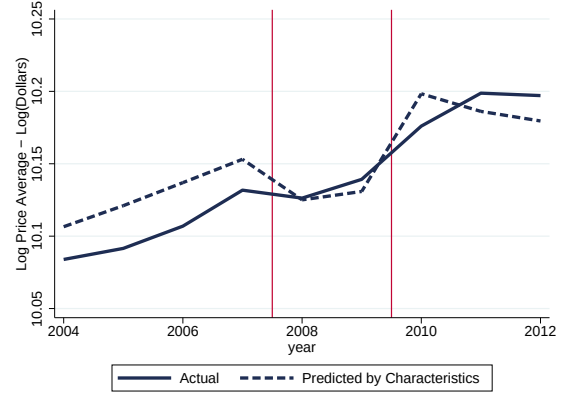
(a) CONSTANT HEDONIC PRICES, ROUNDING



(b) TIME-VARYING HEDONIC PRICES, ROUNDING



(c) CONSTANT HEDONIC PRICES, CONTINUOUS



(d) TIME-VARYING HEDONIC PRICES, CONTINUOUS

Notes: The figure displays the dynamics of average (log) transaction price in the merged Dominion-IHS dataset (solid lines) and the average (log) value predicted with a hedonic regression (dashed lines), when we aggregate both discrete and continuous characteristics of different trims at the model level. Top panels refer to a flexible specification with indicator variables for discrete characteristics, as in equation (1). Within each model, we average the discrete characteristics weighting different trims in proportion to their transaction shares. We then round the average to the closest discrete value, and set the corresponding indicator variable equal to one. Bottom panels refer to an alternative specification that treats all characteristics that vary across trims—including discrete ones—as continuous variables and assumes a log-linear relationship between prices and characteristics. Within each model, we average the discrete characteristics weighting different trims in proportion to their transaction shares and treat the average as the value of a continuous characteristic. Because make and body type do not vary across trims within each model, we control for these two attributes with indicator variables as in equation (1). Left panels refer to constant pre-recession hedonic prices (2004-2007); right panels to time-varying hedonic prices, estimated in three subsamples: pre-recession (2004-2007), recession (2008-2009), and post-recession (2010-2012). Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

Table B1: Hedonic Regressions

PANEL A: COEFFICIENT ESTIMATES OF CONTINUOUS ATTRIBUTES			
	(1)	(2)	(3)
	PRE-RECESSION	RECESSION	POST-RECESSION
LOG(WHEELBASE)	1.138 (0.133)	1.273 (0.168)	1.495 (0.162)
LOG(HORSEPOWER)	0.487 (0.039)	0.488 (0.051)	0.612 (0.051)
LOG(WEIGHT)	0.090 (0.060)	0.153 (0.078)	0.035 (0.077)
LOG(FUEL EFFICIENCY)	-0.080 (0.051)	-0.058 (0.044)	-0.062 (0.047)
LOG(ENGINE SIZE)	0.095 (0.051)	0.028 (0.066)	-0.038 (0.064)
OBSERVATIONS	2,055	1,084	1,671
R^2	0.939	0.958	0.950
PANEL B: QUALITY OF NEW MODELS			
	(1)	(2)	(3)
	DATA	CONSTANT PRICES	TIME-VARYING PRICES
AVERAGE 2008 – AVERAGE 2007	-0.044	-0.052	-0.059
ST. DEV. 2008 – ST. DEV. 2007	-0.073	-0.072	-0.072

Notes: Panel A reports the estimated coefficients of the log of continuous characteristics X_{jt} in equation (1), with standard errors in parentheses, in three subsamples: column (1) refers to the pre-recession subsample (2004–2007); column (2) to the recession subsample (2008–2009); and column (3) to the post-recession subsample (2010–2012). Panel B reports the peak-to-trough dynamics of expenditures and quality of newly introduced models, weighted according to their transaction shares in the IHS dataset. Column (1) reports the difference between the average log price of new models in the 2008 and the average log price of new models in 2007 (first row) and the difference between the standard deviation of log prices of new models in 2008 and the standard deviation of log prices of new models in 2007 (second row). Column (2) reports the difference between the average (first row) and the standard deviation (second row) of predicted log prices, based on constant hedonic prices estimated in the pre-recession subsample, applied to new models introduced in 2008 and to new models introduced in 2007. Column (3) reports the difference between the average (first row) and the standard deviation (second row) of predicted log prices, based on recession hedonic prices applied to new models introduced in 2008 and pre-recession hedonic prices applied to new models introduced in 2007.

models and old models when we aggregate both continuous and discrete characteristics of different trims of each model, using their transaction shares in the IHS dataset, consistent with the hedonic analysis displayed in Figure B12. Our results are robust to this different level of aggregation of car characteristics, buttressing our argument that the level of

Table B2: Hedonic Regressions: New Models

	(1)	(2)	(3)
	PRE-RECESSION	RECESSION	POST-RECESSION
LOG(WHEELBASE)	1.375 (0.272)	2.456 (0.456)	1.815 (0.511)
LOG(HORSEPOWER)	0.409 (0.084)	0.324 (0.142)	0.483 (0.112)
LOG(WEIGHT)	0.340 (0.143)	0.933 (0.205)	0.554 (0.216)
LOG(FUEL EFFICIENCY)	0.051 (0.083)	0.307 (0.118)	-0.253 (0.131)
LOG(ENGINE SIZE)	0.006 (0.104)	-0.288 (0.166)	-0.135 (0.130)
OBSERVATIONS	457	215	306
R^2	0.965	0.969	0.982

Notes: The table reports the estimated coefficients of the log of continuous characteristics X_{jt} in equation (1), with standard errors in parentheses, using data on new models only in three subsamples: column (1) refers to the pre-recession subsample (2004–2007); column (2) to the recession subsample (2008–2009); and column (3) to the post-recession subsample (2010–2012).

aggregation of car characteristics and the exact number of models do not affect our results.

We also perform our estimation of the evolution of the technology frontier dividing carmakers by their geographical origin (Europe, Asia, and US). This strategy is useful for two main reasons. First, these groups of manufacturers are vertically differentiated in terms of average vehicle quality in the US market, as our hedonic-regression analysis confirms. Notably, European manufacturers specialize in higher-quality models. Second, these groups of manufacturers were likely differentially affected by the financial crisis. Specifically, US manufacturers were hit most directly by the crisis, which led to government bailouts.

Exploiting this heterogeneity, we find that European carmakers played a crucial role for the aggregate downward adjustment in the level of technology of new models that we discuss in Section 3.4. Notably, Figure B14 displays the estimates of the year fixed effects in the regression equation (2), which we estimate separately for European, Asian, and US carmakers. The level of technology of new models declined for all three groups during the recession, but European carmakers experienced the largest decline. This evidence suggests that the financial shock hitting US manufacturers is not a primary driver of the overall downward quality adjustment that we document.

We further estimate the level of embodied technology separately for new model names

and new generations of existing model names. We find that the dynamics of average quality are similar for these two groups of models in all periods, including the quality drop during the recession. These results suggest that vertical and horizontal innovations contribute similarly to aggregate quality growth. We display the results of this new analysis in Figure B15.

Figure B16 displays the results we obtain by estimating regression equation (2) without an interaction term between year fixed effects and the indicator function for new models, without sales weights (left panel) and with sales weights (right panel). In this analysis, we effectively pool all models to estimate a common level of technology, and still find a substantial decrease in quality during the Great Recession.

Finally, we perform a back-of-the-envelope calculation of the effects of low new-product quality during the recession for the quality of the overall stock of registered cars in the US. First, we leverage the estimates displayed in the right panel of Figure B16 to obtain a measure of the average annual growth rate in new-car quality, x , during 2004-2007, as well as the average technological level of new cars sold in year t , q_t^N . We normalize $q_{2004}^N = 1$ and assume that the economy in 2004 is on a balanced-growth path with constant inflow of new cars and constant growth in new-car quality equal to x . Thus, in 2004 the quality of cars of age a is given by $q_{a,2004} = (1 + x)^{-a}$. Between 2005 and 2012, we combine these assumptions with our estimates of new-car quality and update the quality of cars of age a as follows: $q_{0,t} = q_t^N$ and $q_{a,t} = q_{a-1,t-1}$ for $a > 0$.

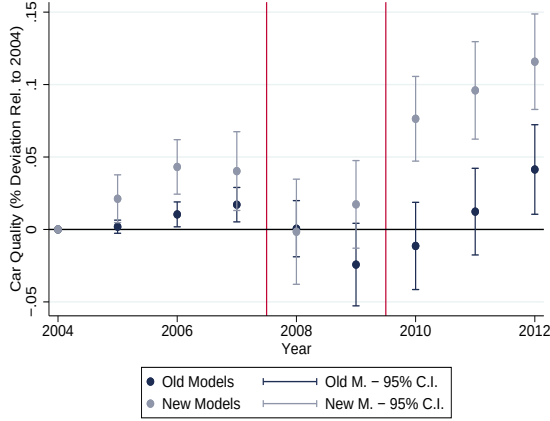
Second, we obtain data on new-vehicle registrations (automobiles and light trucks) from the Bureau of Economic Analysis during 2004-2012 (US Bureau of Economic Analysis, 2022). We assume that vehicles are scrapped at age $a = 15$ (our main findings are similar in a range of values for this parameter) and that the initial age distribution of vehicles is uniform. We update the distribution of vehicle age during 2005-2012 as follows. Let $n_{a,t}$ be the number of cars of age a in year t . We set $n_{0,t}$ equal to the empirical flow of new registrations in year t and $n_{a,t} = n_{a-1,t-1}$ for $a = 1, \dots, 14$.

We then obtain the average quality of the stock, q_t , as follows: $q_t = \frac{\sum_{a=0}^{14} n_{a,t} q_{a,t}}{\sum_{a=0}^{14} n_{a,t}}$.

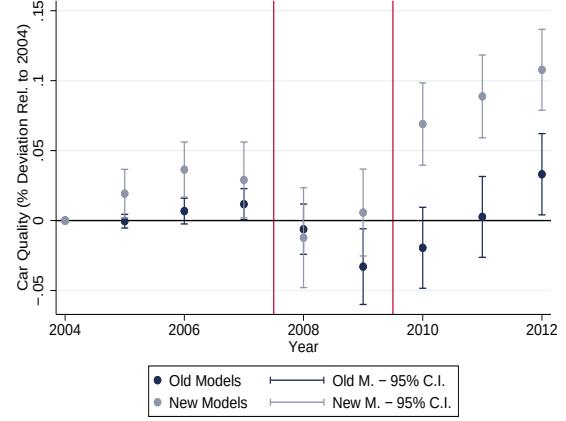
Figure B17 displays the time-series of q_t . We find that at the end of the sample the estimated quality of the stock (solid line) is 1.3% lower than if new-car quality and new-car sales had remained on their pre-recession trend (dashed-dotted line). We further decompose the difference between the estimated quality of the stock and its pre-recession trend in its two components—i.e., changes in the volume of new-car sales and changes in new-product quality. Specifically, the dashed line assumes that new-product quality q_t^N remains on its pre-recession trend, whereas new-car sales follow their empirical path, dropping during the

recession. As the figure shows, this counterfactual scenario accounts for approximately 0.4 percentage points of the overall decline in quality of the stock at the end of the sample, and thus almost one percentage point of the decline is due to the endogenous drop in new-product quality q_t^N .

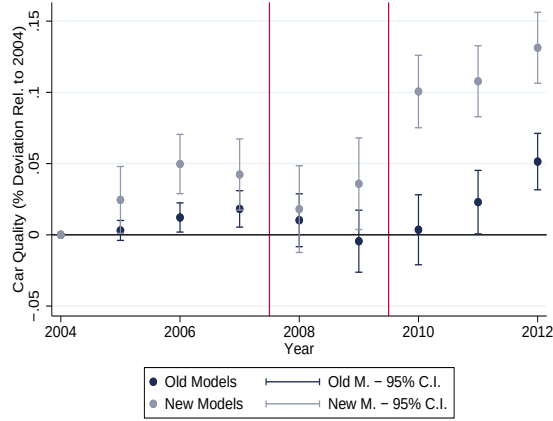
Figure B13: Technology of New and Old Models: Robustness



(a) REPLACING WEIGHT WITH GEOMETRIC DIMENSIONS



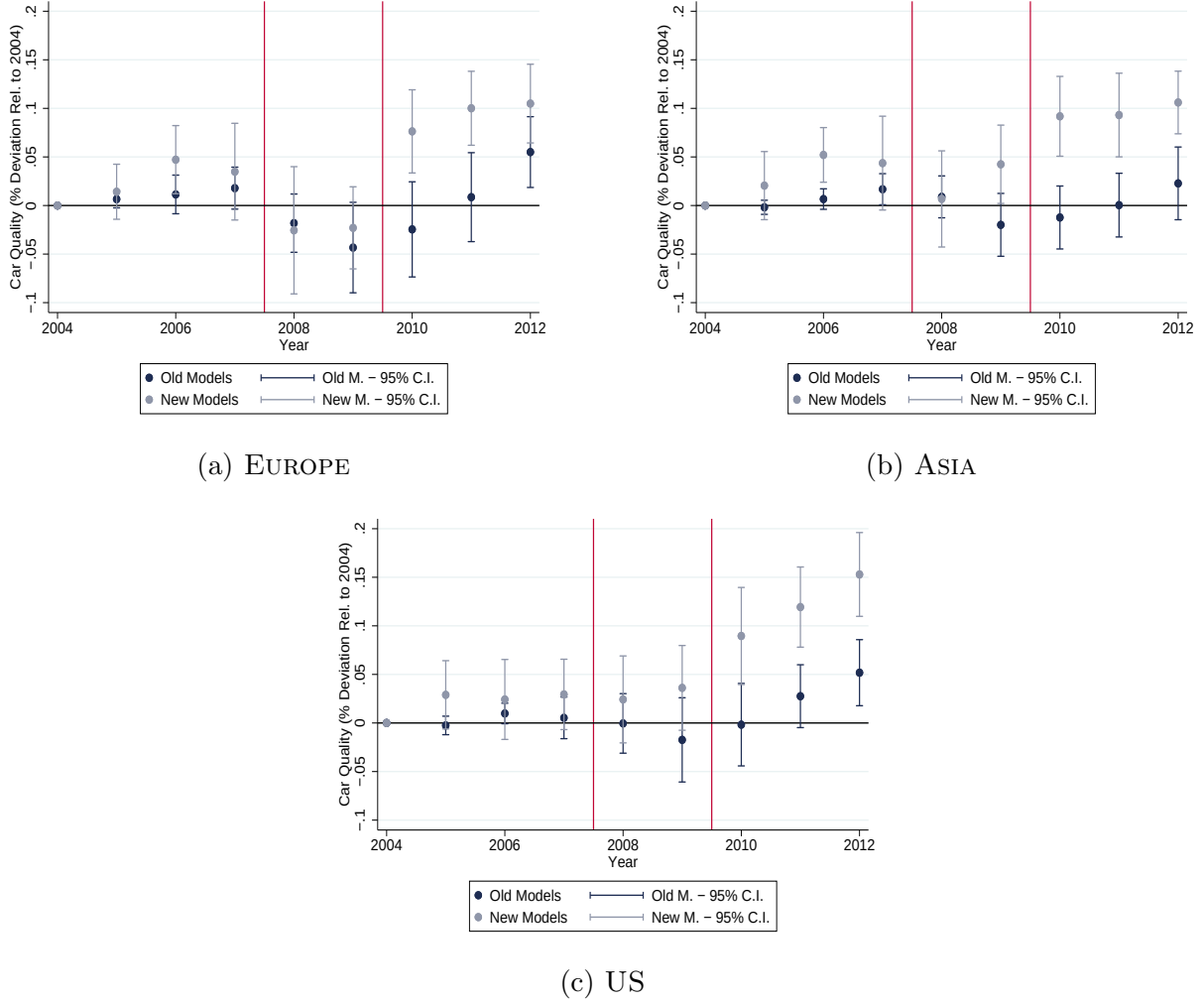
(b) TRANSLOG COST FUNCTION



(c) AGGREGATING ALL CHARACTERISTICS

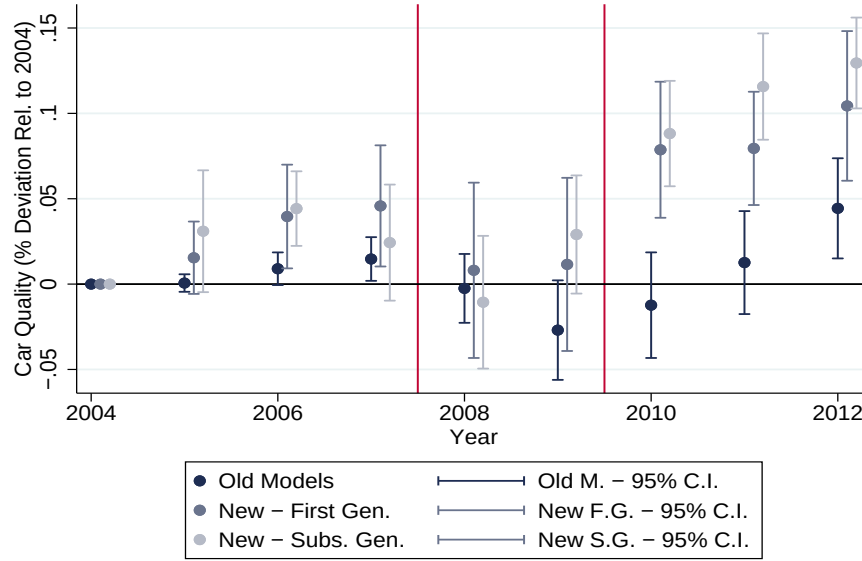
Notes: The figure displays several robustness checks of our measure of technology level for new and old models. Specifically, the top-left panel displays the results we obtain by replacing the variable weight in equation (2) with the variables wheelbase, width, and height. The top-right panel displays the estimates we obtain for regression equation (B3)—i.e., assuming a translog cost function. The bottom panel displays the estimates we obtain when we aggregate both continuous and discrete characteristics of different trims of each model using their transaction shares. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

Figure B14: Technology of New and Old Models by Origin of Carmakers



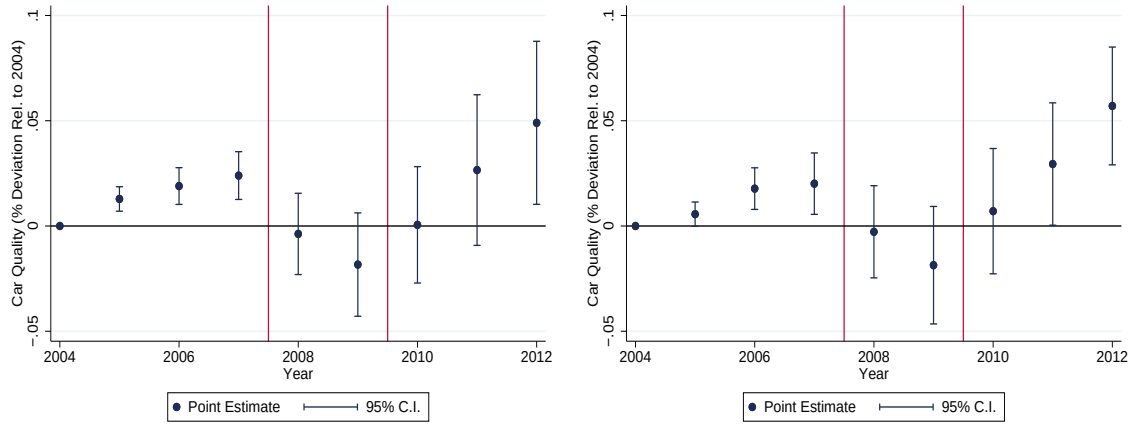
Notes: The figure displays the estimated average level of technological efficiency for new models (clear markers) and old models (dark markers), measured as the estimated time fixed effects in regression equation (2). The left panel refers to European carmakers; the middle panel to Asian carmakers; and the right panel to US carmakers. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

Figure B15: Technology of New Model Names and New Generations



Notes: The figure displays the estimated average level of technological efficiency for new generations of existing model names (clear markers), new model names (intermediate-darkness markers), and continuing models (dark markers). Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

Figure B16: Technology of All Models

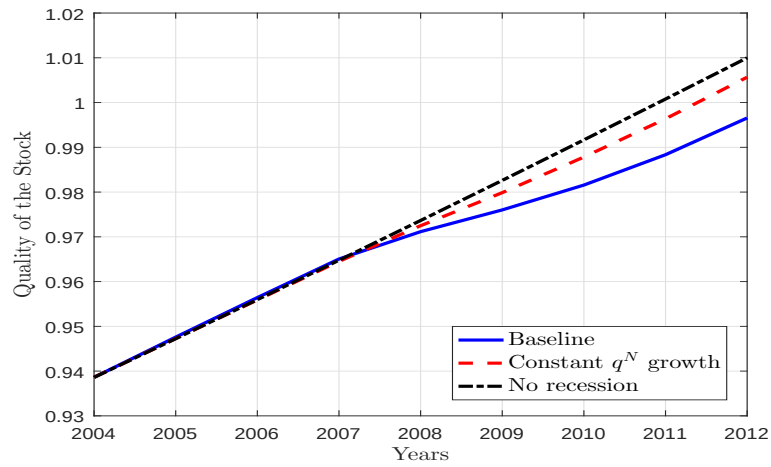


(a) WEIGHTED BY SALES

(b) NOT WEIGHTED BY SALES

Notes: The figure displays the estimated average level of technological efficiency for all models, based on equation (2), removing the interaction term between new models and time. The left panel refers to a regression with weights based on the number of transactions in the IHS dataset, whereas the right panel refers to a regression without weights. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).

Figure B17: Evolution of the Quality of the Stock of Cars



Notes: The figure displays the results of our back-of-the-envelope calculations for the quality of the stock of registered cars, relative to the quality of new cars in 2004. The solid blue line refers to the quality of the stock q_t ; the dashed red line refers to a counterfactual scenario with a constant growth in the quality of new cars, but the empirical inflow of new cars; the dashed-dotted black line refers to a counterfactual with a constant growth in the quality of new cars and a constant inflow of new cars. Horizontal axes report years (2004-2012); vertical lines highlight recession years (2008 and 2009).